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**Knowledge Graph Curation,
Extension and Graph Neural
Network-based Knowledge
Discovery in Lung Cancer Data**

Author: Domenic Schmidt

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Author: Domenic Schmidt

Supervisor:

Antonio Jesús Díaz Honrubia
Departamento de Lenguajes y
Sistemas Informáticos e Ingeniería de
Software
ETSI Informáticos
Universidad Politécnica de Madrid

Co-supervisor:

Paloma Tejera Nevado
Dr. rer. nat.
Centro de Tecnología Biomédica
Universidad Politécnica de Madrid

Abstract

Knowledge graphs have become an important tool for integrating heterogeneous biomedical data, enabling researchers to explore complex relationships across molecular, clinical and environmental domains. The LUCIA knowledge graph, developed at the Universidad Politécnica de Madrid as part of a Horizon Europe research initiative, brings together data on lung cancer from multiple sources within a shared ontological framework. It supports the broader goal of the LUCIA project, which builds a toolbox for discovering new lung cancer risk factors across personal, external and biological dimensions. However, the environmental portion of the graph had accumulated structural problems that prevented meaningful analysis: missing temporal references, bundled multi-year measurements, incorrect entity typing and inconsistent identifiers.

This thesis addresses two connected problems. The first is the curation and extension of the environmental data in the LUCIA knowledge graph. Four automated data pipelines were implemented for the European Environment Agency (EEA), the European Cancer Information System (ECIS), the Organisation for Economic Co-operation and Development (OECD) and the Centers for Disease Control and Prevention (CDC). All data was aligned with the Lung-CABO ontology through a documented set of rules ensuring consistent entity typing, URI conventions and cross-source integration.

The second part applies Graph Neural Network-based link prediction to the extended knowledge graph. A heterogeneous graph was constructed, preserving quantitative features such as pollutant concentrations, cancer mortality rates and gene-disease association scores as node attributes. Three models were compared: Dot Product, TransE and Relational Graph Convolutional Networks (R-GCN). R-GCN achieved the best overall performance and produced biologically plausible novel gene-disease predictions, with established lung cancer genes such as EGFR, ALK, PIK3CA and HIF1A appearing consistently across multiple training runs.

The environmental link predictions represent the central findings of this thesis. The trained R-GCN learned meaningful chemical-disease associations in the embedding space, with heavy metal pollutants (cadmium, arsenic, nickel, lead). The model generated novel chemical-region predictions, identifying cities across Central and Eastern Europe where specific pollutants are predicted to be present but have not been measured. This thesis' results aim to demonstrate that GNN-based link prediction can extract actionable environmental health insights from a curated knowledge graph.

Abbreviations

WHO - World Health Organization
PM2.5 - Particulate Matter (diameter ≤ 2.5 micrometres)
NO₂ - Nitrogen Dioxide
IARC - International Agency for Research on Cancer
EEA - European Environment Agency
ECIS - European Cancer Information System
CDC - Centers for Disease Control and Prevention
OECD - Organisation for Economic Co-operation and Development
KG - Knowledge Graph
GNN - Graph Neural Network
R-GCN - Relational Graph Convolutional Network
TransE - Translating Embeddings (model name)
MRR - Mean Reciprocal Rank
Hits@k - Hits at rank k
SO - Specific Objective
RQ - Research question
EGFR - Epidermal Growth Factor Receptor
ALK - Anaplastic Lymphoma Kinase
PIK3CA - Phosphatidylinositol-4,5-Bisphosphate 3-Kinase Catalytic Subunit Alpha
HIF1A - Hypoxia-Inducible Factor 1-alpha
CLA - ChemicalLocationAssociation
RDF - Resource Description Framework
TTL - Terse RDF Triple Language (Turtle)
SPARQL - SPARQL Protocol and RDF Query Language
URI - Uniform Resource Identifier
SCLC - Small Cell Lung Carcinoma
NSCLC - Non-Small Cell Lung Carcinoma
BaP - Benzo[a]pyrene
C₆H₆ - Benzene
O₃ - Ozone
DNA - Deoxyribonucleic Acid
ROS - Reactive Oxygen Species
RDFS - RDF Schema
OWL - Web Ontology Language
W3C - World Wide Web Consortium
UMLS - Unified Medical Language System
SIO - SemanticScience Integrated Ontology
LOT - Linked Open Terms
Lung-CABO - Lung Cancer Concepts Association Biological Ontology
UPM - Universidad Politécnica de Madrid
LUCIA - Horizon Europe Research Initiative (Grant Agreement 101096473)
GCN - Graph Convolutional Network
GDA - Gene-Disease Association
VDA - Variant-Disease Association
CTD - Comparative Toxicogenomics Database
BIKG - Biological Insights Knowledge Graph
DeePaN - Deep Patient Graph Convolutional Network
NCI/ncit - National Cancer Institute Thesaurus
SNOMED CT - Systematized Nomenclature of Medicine - Clinical Terms
LAU - Local Administrative Unit (Eurostat)
NUTS2 - Nomenclature of Territorial Units for Statistics, Level 2

TL2 - Territorial Level 2
PyG - PyTorch Geometric
DSI - Disease Specificity Index
DPI - Disease Pleiotropy Index
ReLU - Rectified Linear Unit
TPE - Tree-structured Parzen Estimator
Cd - Cadmium
As - Arsenic
Ni - Nickel
Pb - Lead
PM10 - Particulate Matter (diameter ≤ 10 micrometres)
WONDER - Wide-ranging Online Data for Epidemiologic Research
CSV - Comma-Separated Values
GAT - Graph Attention Network
CompGCN - Composition-based Graph Convolutional Network
CAMS - Copernicus Atmosphere Monitoring Service

Table of Contents

1	Introduction.....	1
1.1	Motivation	2
1.2	Problem Statement	2
1.3	Objectives.....	3
1.4	Research questions.....	4
1.5	Thesis structure	4
2	State of the art.....	5
2.1	Lung cancer and environmental risk factors	5
2.2	Knowledge Graphs.....	6
2.2.1	Biomedical Knowledge Graphs	6
2.2.2	The LUCIA Knowledge Graph	7
2.3	Graph Neural Networks	9
2.3.1	Message Passing	9
2.3.2	Graph Convolutional Networks.....	9
2.3.3	Relational Graph Convolutional Networks	10
2.4	Link Prediction on Knowledge Graphs.....	10
2.4.1	Knowledge Graph Embedding Methods	11
2.4.2	Evaluation Protocol	12
2.5	Related Work.....	12
2.5.1	GNNs in Biomedical Knowledge Graphs.....	12
2.5.2	Environmental Data and Knowledge Graphs	13
2.5.3	Prior Work on the LUCIA Knowledge Graph.....	14
3	KG Curation & Extension	15
3.1	Initial Assessment of the Environmental Data.....	15
3.2	Ontological Alignment.....	16
3.3	Data Pipelines	17
3.3.1	EEA Air Quality Pipeline.....	17
3.3.2	ECIS Cancer Statistics Pipeline	18
3.3.3	OECD Expose Pipeline	19
3.3.4	CDC Cancer Statistics Pipeline.....	20
3.4	Shared Entities and Cross-Source Integration	21
3.5	Validation and Summary Statistics.....	21
4	GNN-Based Link Prediction.....	23
4.1	Problem Formulation	23
4.2	Data Preparation	24
4.2.1	Parsing the Environmental Subgraph.....	24
4.2.2	Parsing the Biological Subgraph.....	24
4.2.3	Merging the Two Subgraphs	25

4.2.4	Adding Subtype Bridge Edges.....	25
4.2.5	Output.....	25
4.3	Graph Construction.....	25
4.3.1	Preserving N-ary Reifications.....	26
4.3.2	Node Features.....	27
4.3.3	Building the PyTorch Geometric HeteroData.....	27
4.3.4	Final Graph Statistics	28
4.4	Model Architectures.....	31
4.4.1	Dot Product.....	31
4.4.2	TransE.....	31
4.4.3	R-GCN with DistMult Decoder.....	32
4.5	Training Procedure	33
4.5.1	Data Splits.....	33
4.5.2	Negative Sampling.....	34
4.5.3	Loss Functions.....	34
4.5.4	Optimisation	36
4.5.5	Hyperparameters.....	36
4.6	Evaluation Protocol.....	38
4.6.1	Ranking-Based Evaluation	38
4.6.2	Multi-Seed Evaluation.....	39
4.6.3	Ablation Study	39
4.6.4	Embedding Similarity Test	40
4.6.5	Novel Link Predictions.....	40
5	Results and Discussion.....	42
5.1	Model Comparison.....	42
5.2	The Role of Environmental Data	44
5.3	Cross-Domain Signal Analysis.....	45
5.4	Novel Predictions	47
5.5	Environmental Link Predictions.....	50
5.5.1	Chemical-Disease Associations.....	51
5.5.2	Chemical-Region Predictions	52
6	Conclusions and Future Work.....	55
6.1	Summary of Contributions	55
6.2	Limitations	57
6.3	Future Work.....	58
6.4	Final Remarks	59
7	Bibliography.....	60

List of Tables

Table 3.1: Summary of pipeline outputs	22
Table 4.1: Node types in the Lung-CABO knowledge graph	28
Table 4.2: Summary of the three model architectures	33
Table 4.3: Hyperparameters	36
Table 5.1: Overall link prediction results (mean $\pm$ std across 5 seeds)	42
Table 5.2: GDA-Disease MRR per seed	43
Table 5.3: Ablation study results	44
Table 5.4: Pearson correlation between exposure and mortality	46
Table 5.5: Gene-level stability of novel GDA-Disease predictions	50
Table 5.6: Chemical importance by cosine similarity	51
Table 5.7: Top 20 novel Chemical-Region predictions	53

List of Figures

Figure 2.1: T-box of the biological ontology of the LUCIA knowledge graph	8
Figure 2.2: Previous environmental onotology of the LUCIA KG	9
Figure 3.1: Corrected environmental ontology.....	16
Figure 4.1: Schema of the Lung-CABO knowledge graph	30
Figure 4.2: TransE training loss	35
Figure 4.3: Dot Product training loss	35
Figure 4.4: R-GCN training loss.....	36
Figure 5.1: Overall model comparison bar chart	43
Figure 5.2: Ablation bar chart	45
Figure 5.3: Risk quadrant scatter plot	47
Figure 5.4: Top 20 novel Gene-Disease predictions	49
Figure 5.5: Chemical-Disease embedding similarity heatmap	51

1 Introduction

This chapter introduces the thesis. It presents the motivation behind the work, states the two problems it addresses, defines the objectives and research questions, and outlines the structure of the remaining chapters. Air pollution is one of the leading environmental health risks worldwide. The World Health Organization (WHO) estimates that outdoor air pollution contributes to over four million premature deaths each year, with lung cancer being one of the most significant outcomes [1]. Substances such as fine particulate matter (PM_{2.5}), nitrogen dioxide (NO₂) and benzene have been classified carcinogenic to humans by the International Agency for Research on Cancer (IARC) [2]. Unlike genetic factors, which are tied to individual biology, environmental exposures affect entire populations and regions. This makes them especially relevant for public health policy and prevention strategies.

However, despite its relevance, the relationship between specific environmental exposures and lung cancer subtypes is still not well understood. One reason for this is that the necessary data is scattered across many different sources. Air quality measurements, for example, are collected by agencies like the European Environment Agency (EEA). Cancer incidence and mortality statistics come from registries like the European Cancer Information System (ECIS) and the Centers for Disease Control and Prevention (CDC). Population-level exposure estimations are published by the Organization for Economic Co-operation and Development (OECD). Each of these sources uses its own data formats, geographic granularity and temporal coverage. Bringing them together in a meaningful way is a non-trivial data engineering challenge.

Knowledge graphs offer a natural solution to this problem. A knowledge graph organizes information as a network of entities connected by typed relationships, making it possible to integrate data from very different domains into a single, queryable structure [3]. In the biomedical space, knowledge graphs have been used to combine genomic, clinical and pharmacological data, enabling researchers to explore complex relationships that would be difficult to see in isolated databases [4]. The LUCIA knowledge graph, which was developed at the Universidad Politécnica de Madrid [5], follows this approach for lung cancer. The knowledge graph is one component of the broader LUCIA project, a Horizon Europe research initiative (Grant Agreement 101096473) that aims to improve the understanding of lung cancer risk factors and to support earlier detection. The project analyses three complementary types of risk factors that feed into each other. Personal risk factors cover a person's exposure to chemical pollutants and individual lifestyle and behaviour. External risk factors cover the surrounding environment, including urban, built and transport settings, green spaces, social conditions and climate. Biological responses cover the changes these factors trigger in genetics, epigenetics, metabolism and aging. It brings together molecular, clinical and environmental data within a shared ontological framework. However, the environmental portion of the graph had not been maintained recently and needed to be updated, coverage extended and entities to be typed and identified consistently.

Once a knowledge graph has been populated with high-quality data, it is possible to use machine learning techniques to discover new knowledge from its structure. Graph neural networks (GNNs) are particularly well-suited for this task. They learn representations of entities by aggregating information from neighbouring entities in the graph, capturing both local patterns and broader structural context [6]. When applied to a knowledge graph, GNNs can perform link prediction, scoring the likelihood of a relationship between two entities even

if that relationship is not explicitly recorded. This approach has been used in biomedicine to predict drug-gene interactions, identify resistance mechanisms and discover potential drug repurposing candidates [7].

This thesis combines both of these aspects. The first part focuses on curating and extending the environmental data in the LUCIA knowledge graph, fixing existing issues and adding new data from different sources. The second part of the study applies GNN-based link prediction, specifically using relational GNN architectures along with established knowledge graph embeddings. The aim of this is to explore whether the extended environmental data can reveal new associations with lung cancer.

1.1 Motivation

The starting point for this work was from a pragmatic perspective. When examining the environmental data already present in the LUCIA knowledge graph, it became clear that maintenance was needed before conducting any meaningful analysis. The issues fell into two categories, matching the two types of environmental data in the graph: air quality measurements (Chemical-Location Associations) and cancer statistics (Vital Statistics).

On the air quality side, the EEA data for European cities had correct values but was missing the calendar year for each measurement, making temporal analysis impossible. The OECD data for country-level exposure was more problematic. All PM_{2.5} values across all years, both in micrograms per cubic meter and as percentage of population exposed, were stored in single entity per country. It was impossible to tell which value belonged to which year or which unit of measurement.

On the cancer statistics side, the ECIS data for European registries had demographic information embedded directly in the URI rather than being linked to proper People and Country entities. The CDC data for the United States and Puerto Rico had a similar structural problem: all incidence values across all years were packing into a single entity, with no way to associate a specific year.

Beyond these structural issues, there were also ontological inconsistencies. Entity types were not always correctly assigned, UMLS identifiers were missing or wrong in places. City names contained duplicates caused by encoding differences, accent variations and special characters. All of this meant that the environmental subgraph, while containing valuable information, was not in a state where it could support reliable research.

At the same time, no predictive modelling had been attempted on the environmental portion of the knowledge graph. Prior work within the same research group had successfully applied GNN-based techniques to the molecular and genetic subgraph of LUCIA, focusing on genes, gene fusions and chromosomal rearrangements in relation to lung cancer. That work demonstrated the potential of link prediction on this knowledge graph. The environmental dimension, which covers an entirely different axis of lung cancer risk remained untouched.

These two observations, the need to fix and extend the data and the opportunity to predictive modelling on environmental data, together form the motivation for this thesis.

1.2 Problem Statement

The thesis tackles two connected problems.

The first is a data engineering problem. The environmental data in the LUCIA knowledge graph needed to be restructured, corrected and extended across four data sources. For the EEA data, this meant adding the missing calendar year to each air quality measurement and resolving city-level deduplication issues across 30 European countries. For the OECD data, it required splitting bundled multi-year, multi-unit entities into individual measurements per year and unit. For the ECIS data, demographic information had to be extracted from URIs and linked as proper entities. Historical records going back to 1953 needed to be integrated. For the CDC data, incidence values had to be separated by year and correctly linked to demographic groups. Across all four sources, ontological alignment was necessary to ensure consistent entity typing, identifier schemes and URI conventions.

The second is a machine learning problem. With the corrected and extended knowledge graph in place, the question is whether GNN-based link prediction can discover meaningful new associations between environmental factors and lung cancer subtypes. This requires the construction of a heterogeneous graph that preserves the actual exposure measurements and cancer rates as node features rather than reducing them to binary edges. A key structural challenge is that the environmental data only records one general disease entity (malignant neoplasm of lung), while the biological data contains 42 specific subtypes. Bridging these two layers so that environmental signals can inform subtype-level predictions is a central design problem. Additionally, the graph contains 18 different node types and 24 relation types, each with very different connectivity patterns. This makes it difficult to obtain meaningful overall evaluation metrics and necessitates per-relation analysis.

1.3 Objectives

The **overall goal** of this thesis is to improve the environmental data foundation of the LUCIA lung cancer knowledge graph and to apply GNN-based methods to discover new knowledge from it. This breaks down into five **specific objectives (SO)**:

SO1: Assess the current state of the environmental data in the LUCIA knowledge graph and document all identified gaps, structural problem and inconsistencies across the four data sources (EEA, ECIS, OECD, CDC).

SO2: Design and implement automated data pipelines for each source that produce ontologically compliant RDF output, restructuring broken entities, adding missing temporal references and aligning all data with the LUCIA ontology.

SO3: Validate the extended knowledge graph through statistical checks to ensure completeness, consistency and correctness. This includes creating shared entities (new calendar years, new chemical definitions) used across multiple sources.

SO4: Create a heterogeneous graph representation of the knowledge graph, using node features derived from pollutant exposure values and cancer mortality rates. Train link prediction models using R-GCN as the primary model, with TransE and Dot Product acting as baselines. Connect the biological and environmental layers using disease subtype edges.

SO5: Quantitatively evaluate the results using filtered ranking metrics (MRR and Hits@K) across all relation types. Validate the statistical robustness through multi-seed evaluation and ablation studies. Finally, assess whether the predicted links are biologically plausible.

1.4 Research questions

While the objectives in Section 1.3 describe what this thesis builds, the research questions (RQ) describe what the experiments are meant to answer. Four questions guide the evaluation in Chapter 5.

RQ1: How does the relation-aware R-GCN compare to the shallow embedding baselines, TransE and Dot Product, on link prediction over the full graph? This question tests whether relation-specific message passing adds predictive power beyond simple entity and relation embeddings.

RQ2: Does adding environmental data to the biological subgraph improve gene-disease link prediction? This is the central question behind the ablation study. It compares the combined graph against a biological-only and an environmental-only variant to measure the contribution of the environmental relations.

RQ3: Does the model learn meaningful cross-domain associations between air pollutants and lung cancer in its embedding space? This question looks past the ranking metrics and examines whether the learned representations position known carcinogens close to the disease subtypes they affect.

RQ4: Are the novel link predictions plausible and stable, and what environmental associations does the model surface? This question covers both the biological predictions, assessed for biological plausibility and stability across random seeds, and the environmental predictions, including chemical-disease associations and unmeasured chemical-region exposures.

Chapter 5 answers these four questions in turn. Section 5.1 addresses RQ1, Section 5.2 addresses RQ2, Section 5.3 addresses RQ3 and Sections 5.4 and 5.5 address RQ4.

1.5 Thesis structure

The remainder of this thesis is organised as follows. Chapter 2 reviews the state of the art in knowledge graphs, graph neural networks, link prediction and their applications in biomedical and environmental research. Chapter 3 describes the curation and extension of the environmental data in the LUCIA knowledge graph, covering the data pipelines for each source and the ontological alignment process. Chapter 4 presents the graph construction, the R-GCN model architecture and the experimental setup for link prediction. Chapter 5 discusses the results, including the multi-seed evaluation, ablation study, novel link predictions and cross-domain analysis. Chapter 6 concludes the thesis and outlines directions for future work.

2 State of the art

2.1 Lung cancer and environmental risk factors

This chapter reviews the background and related work. It covers lung cancer and its environmental risk factors, the foundations of knowledge graphs and graph neural networks, the link prediction task and the most relevant prior work, including earlier research on the LUCIA knowledge graph.

Lung cancer is broadly classified into two histological categories: small cell lung carcinoma (SCLC) and non-small cell lung carcinoma (NSCLC). NSCLC accounts for approximately 85% of all diagnoses and includes subtypes such as adenocarcinoma, squamous cell carcinoma and large cell carcinoma. While tobacco smoking remains the dominant risk factor, present in roughly 85% of cases, the role of environmental exposures has received increasing attention over the past two decades [8].

Globally, lung cancer remains the leading cause of cancer-related death, accounting for approximately 1.8 million deaths per year [9]. A major factor in this burden is that a large proportion of cases are diagnosed at advanced stages, when treatment options are limited and survival rates drop significantly. The five-year survival rate remains among the lowest of all cancer types, generally falling between 10% and 20%, depending on the country and stage at diagnosis [9].

In 2013, the International Agency for Research on Cancer (IARC) classified outdoor air pollution, as a whole and fine particulate matter (PM_{2.5}), specifically, as carcinogenic to humans (Group 1) [2]. This classification was based on extensive epidemiological evidence linking long-term exposure to ambient air pollution with elevated lung cancer incidence and mortality. The landmark cohort study by Pope *et al.* [10], which followed over 500,000 adults in the United States across 16 years, found that each 10 micrograms per cubic metre increase in PM_{2.5} concentration was associated with an approximately 8% increase in lung cancer mortality, after adjusting for smoking and other confounders.

Beyond PM_{2.5}, several other airborne pollutants have been linked to lung cancer risk. Nitrogen dioxide (NO₂), a marker of traffic-related air pollution, has shown positive associations with lung cancer incidence in European cohort studies. Benzo[a]pyrene (BaP), a polycyclic aromatic hydrocarbon released during incomplete combustion, is classified as a Group 1 carcinogen by IARC and is known to form DNA adducts that can initiate carcinogenesis. Benzene (C₆H₆), an established cause of haematological malignancies, has also been investigated as a potential contributor to lung cancer risk through chronic, low-level exposure. Ozone (O₃), while not directly classified as a carcinogen, contributes to oxidative stress in lung tissue and may act as a promoter in individuals already exposed to other risk factors.

The biological mechanisms through which air pollutants contribute to lung cancer are not fully understood, but several pathways have been established. PM_{2.5} particles are small enough to penetrate deep into the alveoli, where they trigger chronic inflammation and generate reactive oxygen species (ROS), that cause oxidative damage to DNA. Repeated exposure leads to persistent tissue inflammation, which creates a microenvironment favourable to tumour

development. Recent research by Hill *et al.* [11] has suggested that air pollution may promote lung cancer through an inflammation-mediated mechanism that activates pre-existing, dormant driver mutations, particularly in the Epidermal Growth Factor Receptor (EGFR) gene, rather than by directly causing new mutations. This finding is especially relevant because it offers a mechanistic explanation for why lung cancer also occurs in non-smokers living in polluted environments.

A key challenge for studying these relationships at a population level is that the relevant data is scattered across multiple independent sources. Air quality measurements are collected by monitoring networks such as the European Environment Agency (EEA), which operates thousands of stations across 30 European countries. Cancer incidence and mortality statistics are maintained by registries such as the European Cancer Information System (ECIS), which aggregates data from national cancer registries with varying temporal and geographic coverage. Population-level exposure estimates, which combine monitoring data with atmospheric models to estimate the pollutant concentrations experienced by entire populations, are published by the Organisation for Economic Co-operation and Development (OECD). Each of these sources uses its own data formats, spatial granularity (city-level, regional, national) and temporal resolution. Integrating them into a coherent structure that allows joint analysis is a non-trivial data engineering problem and it is precisely the kind of problem that knowledge graphs are designed to solve.

2.2 Knowledge Graphs

A knowledge graph is a structured representation of real-world information, in which entities are modelled as nodes and the relationships between them as typed, directed edges. Each fact is expressed as a triple of the form (subject, predicate, object). For example, the triple (PM2.5, *exposure_in*, Madrid) states that PM2.5 exposure has been measured in Madrid. The collection of many such interconnected triples forms the graph [3]. While several definitions of knowledge graphs exist in the literature [12], the common elements are the use of a graph-based data model, the representation of both entities and their semantic types and the ability to integrate information from heterogeneous sources into a single, queryable structure.

Knowledge graphs are typically built on top of formal ontologies that define the vocabulary of entity types, relationship types and constraints. The Resource Description Framework (RDF) is the World Wide Web Consortium (W3C) standard for encoding knowledge graphs as sets of triples, where entities are identified by Uniform Resource Identifiers (URIs) and relationships are defined by ontological properties [13]. RDF Schema (RDFS) extends RDF with basic vocabulary for defining class hierarchies and property domains [14]. The Web Ontology Language (OWL) adds richer semantics on top of RDFS, enabling property restrictions, cardinality constraints and logical axioms [15]. SPARQL is the standard query language for retrieving and manipulating data stored in RDF format [16].

2.2.1 Biomedical Knowledge Graphs

In biomedicine, knowledge graphs have become an important tool for integrating heterogeneous data from different domains. The need for integration arises because biomedical knowledge is distributed across hundreds of specialised databases, each covering a different aspect of biology or medicine and using its own identifiers, schemas and formats [4]. Knowledge graphs provide a way to

link these resources into a single structure, where relationships across domains become explicit and queryable.

Several large-scale biomedical knowledge graphs have been developed. The Unified Medical Language System (UMLS), maintained by the U.S. National Library of Medicine, integrates over 200 biomedical vocabularies and provides a metathesaurus of concepts and relationships that serves as a reference for terminology mapping across systems [17]. DisGeNET aggregates gene-disease and variant-disease associations from curated databases, animal models and scientific literature, covering over 30,000 diseases and 21,000 genes in its current version [18]. DrugBank combines detailed drug data with comprehensive drug target information, linking pharmacological and molecular data for over 15,000 drug entries [19]. More recently, PrimeKG was constructed specifically for precision medicine applications, integrating 20 biomedical resources into a single heterogeneous graph with over 100,000 nodes and 4 million edges, covering diseases, drugs, genes, proteins, pathways and anatomical entities [20].

These resources have enabled a range of computational applications. Himmelstein *et al.* demonstrated systematic drug repurposing using Hetionet, a heterogeneous graph integrating 29 public resources, by learning to predict new drug-disease treatment relationships from the graph structure [21]. Nicholson and Greene provided a comprehensive review of knowledge graph construction methods and their biomedical applications, covering use cases from drug target identification to adverse effect prediction [4]. More broadly, graph representation learning methods applied to biomedical knowledge graphs have shown promise for predicting protein functions, identifying disease mechanisms and prioritising therapeutic targets [22].

2.2.2 The LUCIA Knowledge Graph

The LUCIA knowledge graph was developed at the Universidad Politécnica de Madrid (UPM) as part of the LUCIA project, a Horizon Europe research initiative (Grant Agreement: 101096473) focused on understanding lung cancer risk factors and improving early detection strategies [5]. The knowledge graph integrates molecular, clinical and environmental data within a unified ontological framework defined by the Lung-CABO ontology (Lung Cancer Concepts Association Biological Ontology) [5]. The ontology was developed following the Linked Open Terms (LOT) methodology [23], an ontology engineering framework also developed at UPM and uses the SemanticScience Integrated Ontology (SIO) [24] as its backbone for entity typing and property definitions.

The biological portion of the graph contains entities representing 42 lung cancer subtypes, organised in a disease hierarchy, genes with their gene-disease association scores sourced from DisGeNET, gene fusions and chromosomal rearrangements from the Mitelman Database and COSMIC, biological pathways from WikiPathways, sequence variants and biomarkers. These are connected through typed relationships, such as *associated_with* (linking genes to diseases), *in_pathway* (linking genes to pathways) and *has_fusion* (linking diseases to gene fusions). Figure 2.1 shows the T-box of the biological ontology.

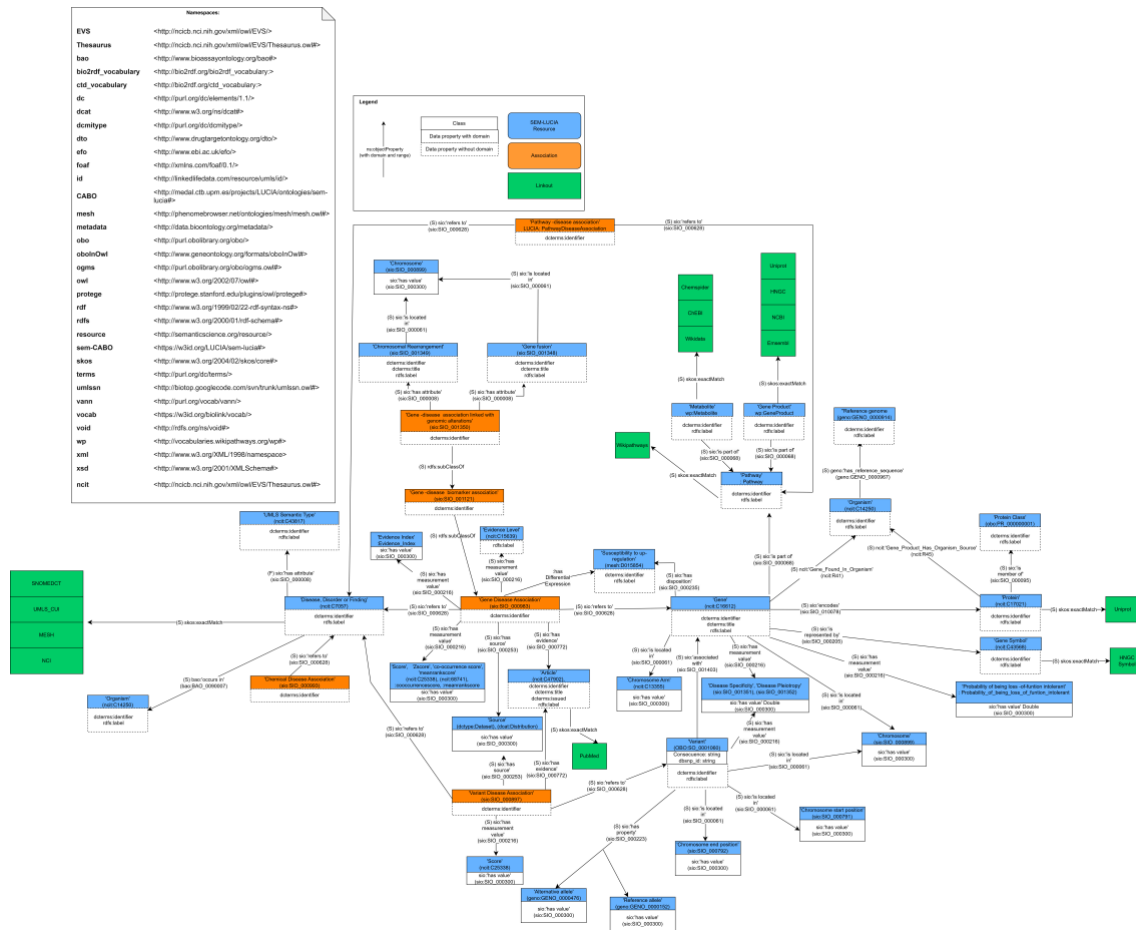


Figure 2.1: T-box of the biological ontology of the LUCIA knowledge graph

The environmental portion of the graph contains air quality measurements from the European Environment Agency (EEA), covering pollutant concentrations for PM2.5, NO2, benzene, benzo[a]pyrene and other substances across thousands of monitoring stations in 30 European countries. Cancer incidence and mortality statistics from the European Cancer Information System (ECIS) provide demographic breakdowns by country, age group and gender. Population-level exposure estimates from the Organisation for Economic Co-operation and Development (OECD) complement the station-level EEA data, with regional aggregates that estimate the pollutant concentrations experienced by entire populations. Figure 2.2 shows the environmental ontology as it existed before the curation work described in Chapter 3.

A key structural property of the graph is its heterogeneity. The full graph contains 18 different node types and 24 different relation types, reflecting the diversity of the underlying data sources. Disease nodes serve as the bridge entities that connect the biological and environmental subgraphs. Through the disease hierarchy (*subtype_of* edges), the single, general disease entity used in the environmental data (malignant neoplasm of lung) is linked to the 42 specific subtypes used in the biological data. This bridge makes it possible, at least in principle, to trace paths from a specific air pollutant through geographic regions and cancer statistics to molecular-level gene associations.

passing framework by defining the message from a neighbour as a linear transformation of that neighbour's features, with normalisation based on the degrees of both the sending and receiving nodes. Concretely, the update rule applies a shared weight matrix W to the neighbourhood features, normalised by the symmetric degree matrix, followed by a non-linear activation function.

GCNs are effective for tasks on homogeneous graphs, where all edges represent the same type of relationship. However, they use a single, shared weight matrix for all edges. This means that a GCN treats every relationship identically, whether it connects a gene to a disease, a chemical to a region, or a pathway to a gene. For multi-relational knowledge graphs, this is a significant limitation.

2.3.3 Relational Graph Convolutional Networks

Relational Graph Convolutional Networks (R-GCNs), proposed by Schlichtkrull *et al.* [28], extend GCNs to handle multiple relation types. The key idea is straightforward: instead of one weight matrix shared across all edges, R-GCN learns a separate weight matrix W_r for each relation type r . During message passing, a node groups its incoming messages by relation type, applies the corresponding weight matrix to each group and sums the results together with a self-connection term.

This means that the model can learn, for example, that a gene connected to a disease through an *associated_with* edge should be treated differently from a chemical connected to a region through an *exposure_in* edge. Each relation type gets its own learned transformation, which allows the model to capture the different semantics and connectivity patterns of each relationship.

One practical problem with this approach is scalability. If the graph has R relation types and the hidden dimension is d , R-GCN needs R separate weight matrices of size $d \times d$, which adds up quickly. To control this, Schlichtkrull *et al.* proposed two regularisation strategies. Basis Decomposition expresses each relation-specific weight matrix as a linear combination of a small number of shared basis matrices: $W_r = \sum (a_{r,b} * B_b)$, where B_b are the basis matrices and $a_{r,b}$ are learnable coefficients. This reduces the number of free parameters from $R * d * d$ to $B * d * d + R * B$, where B is the number of bases and is typically much smaller than R . The alternative, block-diagonal decomposition, restricts each W_r to be block-diagonal, reducing parameters at the cost of limiting interactions between feature dimensions [28].

R-GCN is particularly well suited for the LUCIA knowledge graph. The graph has 24 different relation types, with very different connectivity patterns. Some relations, like *exposure_in*, connect a small number of chemicals to thousands of regions. Others, like *associated_with*, link thousands of genes to a few dozen diseases. A single, shared weight matrix would be forced to compromise across all these patterns. Relation-specific weights allow the model to learn appropriate transformations for each one.

2.4 Link Prediction on Knowledge Graphs

Link prediction is the task of inferring missing relationships in a knowledge graph. Given a set of known triples, the goal is to predict which additional triples are likely to be true but have not been recorded. This is relevant in any knowledge graph because no knowledge graph is complete. In biomedical graphs specifically, link prediction has been used to discover drug-gene interactions, identify drug repurposing candidates and predict disease-gene associations [4].

The general approach works as follows. A scoring function, $f(h, r, t)$, is defined that assigns a plausibility score to any candidate triple, where h is the head entity, r is the relation and t is the tail entity. The model is trained so that known true triples receive high scores and randomly corrupted triples receive low scores. At inference time, the scoring function can be applied to any candidate triple to estimate how likely it is to be true.

2.4.1 Knowledge Graph Embedding Methods

Knowledge Graph Embedding methods learn low-dimensional vector representations for all entities and relations in the graph. The key idea is to map entities and relations into a continuous vector space, where the geometry of the space reflects the structure of the graph. Different methods define different geometric relationships between the embeddings of h , r and t [29].

TransE, introduced by Bordes *et al.* [30], is one of the simplest and most influential embedding methods. It models each relation as a translation in embedding space. If a triple (h, r, t) is true, then the embedding of the head plus the embedding of the relation should be close to the embedding of the tail: $h + r$ should approximately equal t . The scoring function is the negative distance: $f(h, r, t) = -||h + r - t||$. TransE is efficient and works well for one-to-one relations, but it has difficulty with symmetric relations (where $h + r = t$ would require $r = 0$) and with one-to-many relations (where multiple different tail entities would need to map to the same point) [29].

The second baseline is the Dot Product scoring function, defined simply as $f(h, t) = \text{sum}(h * t)$, where $*$ denotes element-wise multiplication. This is the inner product of the head and tail embeddings. It does not use the relation at all. A triple is scored purely based on how similar the head and tail embeddings are in the learned vector space.

The Dot Product has no single origin paper in the knowledge graph embedding literature. It arises as the degenerate case of more expressive bilinear models. DistMult [31] reduces to the Dot Product when its relation vector is set to the all-ones vector. RESCAL [32] reduces to it when its relation matrix is the identity. From a different angle, Dot Product is mathematically equivalent to low-rank factorisation of the adjacency matrix of the graph [33]. In the GNN literature, it is the canonical decoder of graph autoencoders, introduced by Kipf and Welling [34] and the default training objective of GraphSAGE [35]. Open Graph Benchmark uses it as the reference baseline for several link prediction tasks [36].

The reason for including Dot Product as a baseline is methodological. Together with TransE and R-GCN, it forms a three-way comparison that isolates different sources of predictive signal. Dot Product uses only entity embeddings, with no relation modelling. TransE uses translation-based relations. R-GCN uses relation-specific message passing on top of the graph structure. If Dot Product already performs well, then entity embeddings alone carry most of the signal. If R-GCN performs significantly better, then the relation-specific message passing is doing real work. Empirical studies have shown that such minimal baselines, when properly trained, are often competitive with more elaborate models [37]. This makes Dot Product a strong baseline, rather than a trivial one.

Both TransE and Dot Product are shallow embedding methods. They treat each triple independently, without considering the broader graph neighbourhood. This is the fundamental difference compared to GNN-based approaches like R-

GCN, which aggregate information from neighbouring nodes through message passing before scoring a candidate triple [28].

2.4.2 Evaluation Protocol

Link prediction models are evaluated using a ranking-based protocol that was introduced alongside the TransE model [30]. For each test triple (h, r, t), the tail entity t is replaced by every entity in the graph and the scoring function ranks all candidates. The rank of the correct entity is recorded. The same procedure is repeated for head prediction, replacing h with every entity. The final metrics are averaged over both directions.

Two standard metrics are reported. Mean Reciprocal Rank (MRR) is the average of the reciprocal ranks across all test triples. If the correct entity is ranked first, the reciprocal rank is 1. If it is ranked fifth, the reciprocal rank is 0.2. Higher MRR indicates better ranking quality. Hits@k measures the proportion of test triples where the correct entity appears in the top k predictions. Hits@1, Hits@3 and Hits@10 are commonly reported. On a graph with thousands of entities, even a Hits@10 of 0.5 is a strong result because it means the model places the correct answer in the top 10 out of thousands of candidates in half of all cases.

The standard evaluation uses the filtered setting, also introduced by Bordes *et al.* [30]. During ranking, other known true triples are removed from the candidate set, so that valid but unobserved triples are not incorrectly penalised. For example, if Gene_X is associated with both Disease_A and Disease_B and the test triple is (Gene_X, *associated_with*, Disease_A), then Disease_B is removed from the ranking candidates. Without this filtering, the model would be penalised for correctly ranking Disease_B highly. The filtered setting is now the universally accepted standard in the knowledge graph embedding literature [38].

An important consideration when interpreting these metrics is that overall numbers can be misleading on heterogeneous graphs. A graph like the LUCIA knowledge graph has relation types with very different characteristics. Some relations connect a few source entities to thousands of targets, making them easy to predict. Others connect thousands of sources to a few dozen targets, which is harder. Reporting only overall MRR can hide poor performance on the most interesting relations behind high performance on trivial ones. For this reason, per-relation evaluation is essential [38].

2.5 Related Work

The application of graph-based machine learning to biomedical knowledge graphs has grown rapidly in recent years. This section reviews the most relevant work across three areas: GNNs applied to biomedical knowledge graphs, with a focus on lung cancer, the gap at the intersection of environmental data and knowledge graph-based prediction and prior work on the LUCIA knowledge graph itself.

2.5.1 GNNs in Biomedical Knowledge Graphs

Zitnik *et al.* introduced Decagon [7], a graph convolutional model for predicting polypharmacy side effects. The model operated on a multimodal graph combining protein-protein interactions with drug-drug interactions, where each side effect was represented as a separate edge type. Decagon was among the first models to demonstrate that GNNs can perform multi-relational link prediction in biomedicine, outperforming conventional baselines by a wide

margin. The architectural idea of learning relation-specific transformations on a heterogeneous biomedical graph directly influenced later work, including the R-GCN approach used in this thesis.

In the specific context of lung cancer, several graph-based approaches have been proposed. Gogleva *et al.* [39] constructed a large-scale heterogeneous knowledge graph integrating 37 data sources covering clinical, preclinical and bibliographic information. Their recommendation framework prioritised genes potentially involved in resistance to targeted therapies against EGFR-mutant non-small cell lung cancer. The system identified 57 resistance markers among more than 3,000 candidate genes, including mechanisms that had not been reported before. This work showed that knowledge graph-based methods can surface biologically actionable predictions even when the underlying graph is noisy and incomplete.

Fang *et al.* [40] took a different approach by integrating the AstraZeneca Biological Insights Knowledge Graph (BIKG) into survival models for NSCLC patients from two atezolizumab clinical trials (POPLAR and OAK). They showed that features derived from the knowledge graph produced significantly improved hazard ratios compared to tumour mutation burden alone. This was the first rigorous demonstration that knowledge graph priors translate into measurable clinical gains in registrational-trial data.

On the patient-level side, Fang *et al.* [41] developed DeePaN, a graph convolutional model that builds patient similarity graphs from clinico-genomic features. Applied to 1,937 immunotherapy-treated NSCLC patients, DeePaN identified five subgroups with significantly different survival outcomes, that non-graph methods failed to separate. More recently, Vaida *et al.* [42] integrated metabolomic and demographic data into a GNN framework for early-stage lung cancer detection, achieving 89% test accuracy. While focused on classification rather than link prediction, this work demonstrated that combining heterogeneous data modalities within a graph structure adds predictive value beyond what any single data type provides.

2.5.2 Environmental Data and Knowledge Graphs

A notable gap exists at the intersection of environmental exposure data and knowledge graph-based disease prediction. The Comparative Toxicogenomics Database (CTD) [43] curates over 45 million toxicogenomic relationships, linking chemicals to genes, phenotypes and diseases. Its chemical-gene-phenotype-disease tetramer engine can trace mechanistic pathways from specific air pollutants through biological intermediates to disease outcomes. However, CTD has primarily been used for toxicological research, rather than for graph-based predictive modelling.

In the air quality domain, GNNs have been applied to pollution forecasting. Li *et al.* [44] encoded physical constraints into a graph convolutional model for multi-pollutant prediction, improving extrapolation accuracy by 11-22% over baselines on a China-wide monitoring network. These models produce the spatiotemporal exposure estimates that could, in principle, serve as input features for a disease prediction model. Yet, no published work has connected these two threads: using GNN-estimated or measured pollution exposures as node features within a biomedical knowledge graph for disease-related link prediction.

This gap is precisely what this thesis addresses. The LUCIA knowledge graph is the first knowledge graph that integrates station-level air quality measurements

(EEA), regional exposure estimates (OECD) and cancer registry statistics (ECIS) alongside molecular and genetic data within a single graph structure. The link prediction experiments in this thesis represent the first attempt to evaluate whether this environmental data can improve gene-disease predictions when combined with biological information through a GNN.

2.5.3 Prior Work on the LUCIA Knowledge Graph

Prior work conducted within the same research group at UPM applied a hybrid neuro-symbolic approach to the molecular and genetic subgraph of the LUCIA knowledge graph. That work combined AnyBURL [45], a bottom-up rule learning system that extracts logical rules from knowledge graphs via random walks, with a GCNConv-based graph neural network. The focus was exclusively on the biological dimension of lung cancer: genes, gene fusions, chromosomal rearrangements and their associations with disease subtypes. The hybrid model produced interpretable predictions through the extracted rule paths and demonstrated that the LUCIA graph structure supports meaningful link prediction.

The present thesis differs from that work in three important ways. First, it works with both the biological and environmental subgraphs simultaneously, constructing a combined heterogeneous graph with 18 node types and 24 relation types. Second, it uses R-GCN with basis decomposition instead of standard GCNConv. This is a better fit for the multi-relational structure of the full graph because R-GCN learns separate transformation weights for each relation type, rather than treating all edges identically. Third, this thesis incorporates node features derived from actual pollutant concentrations and cancer mortality rates, rather than relying solely on graph structure. The inclusion of these features allows the model to use quantitative environmental measurements, not just the fact that an edge exists.

Existing work has established that GNNs can predict gene-disease associations in molecular knowledge graphs, that knowledge graph priors improve clinical predictions in lung cancer and that environmental factors are relevant risk factors for the disease. However, no prior study has combined environmental exposure data with biological data within a single, heterogeneous knowledge graph for link prediction. The central question of this thesis is whether the environmental data provides measurable improvements in gene-disease link prediction performance. The ablation study presented in Chapter 5, comparing the combined graph against a biological-only and environmental-only graph, is designed to answer exactly this question.

3 KG Curation & Extension

This chapter describes the curation and extension of the environmental data in the LUCIA knowledge graph. It presents the initial assessment of the existing data, the ontological alignment rules, the four data pipelines and the validation steps applied to the generated output.

3.1 Initial Assessment of the Environmental Data

The starting point for the knowledge graph curation work was a systematic assessment of the environmental data already present in the LUCIA knowledge graph. This assessment was guided by a reference document that identified the specific problems requiring resolution. The issues affected both types of environmental data in the graph: air quality measurements, stored as Chemical-Location Association (CLA) entities, and cancer statistics, stored as Vital Statistics entities.

The environmental data originated from four sources. The European Environment Agency (EEA) provided city-level air quality measurements for European cities. The Organisation for Economic Co-operation and Development (OECD) provided country-level and regional pollutant exposure estimates. The European Cancer Information System (ECIS) provided lung cancer incidence and mortality rates for European countries and registries. The Centers for Disease Control and Prevention (CDC) provided lung cancer incidence data for the United States and Puerto Rico.

Each source had distinct structural problems. The EEA data contained one CLA instance per measurement, each linking a chemical to a European city with a concentration value. The values themselves were correct, but the calendar year for each measurement was missing, making it impossible to distinguish measurements from different years. The OECD data had a more serious problem: for each country, all PM2.5 values across all years and both units of measurement were stored in a single CLA entity, with no way to determine which value corresponded to which year or unit. The ECIS data stored cancer statistics as Vital Statistics entities, but the demographic information (country, age group, gender) was encoded directly in the URI string rather than being linked to proper People and Country entities through typed relationships, so any SPARQL query filtering by country or demographic group would fail. The CDC data had a problem similar to the OECD data: all incidence values across all years were packed into a single Vital Statistics entity per demographic group.

Beyond these source-specific problems, the initial assessment revealed several ontological inconsistencies that affected all sources. Entity types were not always correctly assigned according to the SIO and NCI Thesaurus vocabularies used by the LUCIA ontology. UMLS identifiers were missing or incorrect for some chemical entities. City names contained duplicates caused by encoding differences, accent variations and inconsistent use of special characters. Some entities that already existed in the knowledge graph were being redefined in the new TTL files rather than being referenced, which would create duplicate descriptions and potential conflicts when loaded into the triplestore.

3.2 Ontological Alignment

Before implementing the data pipelines, a set of ontological rules had to be established. These rules ensured that all new data would be consistent¹ with the existing LUCIA knowledge graph and with the Lung-CABO ontology. The alignment was developed in two stages. An initial set of corrections was defined based on the assessment described in Section 3.1. After review of the generated TTL files against the ontological specification, a second set of corrections refined entity typing, URI conventions and the handling of shared entities. Figure 3.1 shows the corrected environmental ontology that results from these rules.

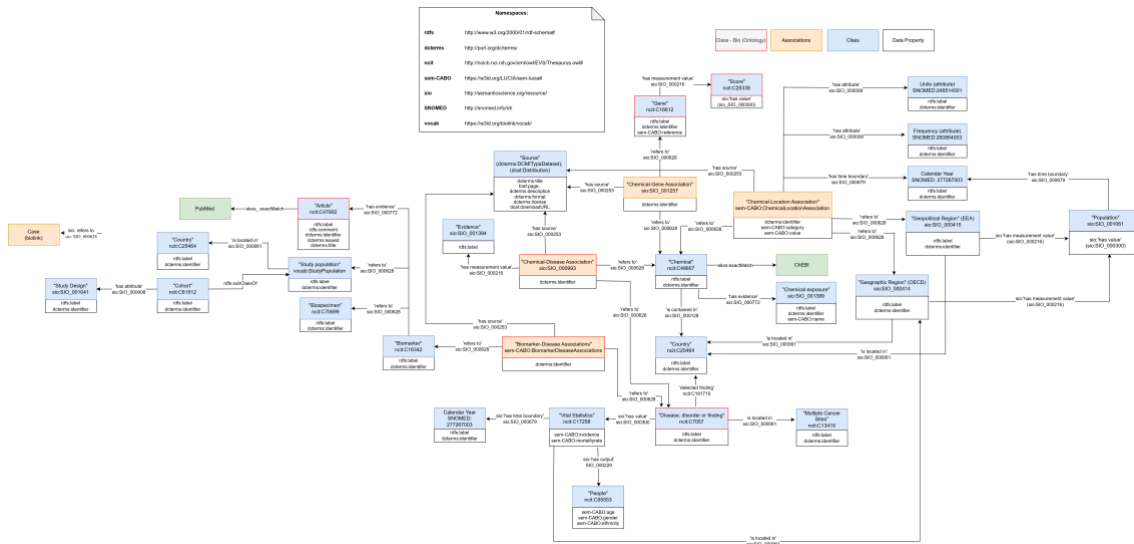


Figure 3.1: Corrected environmental ontology

The most important type corrections were as follows. Country entities use `ncit:C25464` (the original data had some countries typed incorrectly as `ncit:C61066`). City-level entities from the EEA data are typed as `Geopolitical Regions (sio:SIO_000415)`, reflecting that they represent administrative units at the city level. Regional entities from the OECD data are typed as `Geographic Regions (sio:SIO_000414)`, reflecting that they represent statistical regions at the NUTS2/TL2 level. This distinction between `SIO_000415` and `SIO_000414` preserves the different spatial granularities of the two data sources. CalendarYear entities use the SNOMED CT concept `277267003`, rather than the SKOS Concept type that had been used in an earlier version.

Chemical entities in the LUCIA ontology are identified by their UMLS Concept Unique Identifiers. The EEA data covers ten pollutants, five of which already existed in the ontology: `PM2.5 (C5890534)`, `benzo[a]pyrene (C0005052)`, `benzene (C0005036)`, `nitrogen dioxide (C0028160)` and `ozone (C0030106)`. These existing chemicals are only referenced by their URI and never redefined. The remaining five are `PM10 variants` that had to be created as new entities: `PM10 (C1720884_10)`, `PM10 with arsenic (C1720884_10_As)`, `PM10 with cadmium (C1720884_10_Cd)`, `PM10 with nickel (C1720884_10_Ni)` and `PM10 with lead (C1720884_10_Pb)`.

A consistent URI structure was defined for all entity types, with the base namespace `http://medal.ctb.upm.es/projects/LUCIA/res/sem-lucia#`.

¹ See `shared/ttl_utils.py`.

Geopolitical Regions follow the pattern country/gpr/<CountryCode><CityName>. Geographic Regions follow country/gr/<CountryCode><RegionName>. Vital Statistics follow vitalstatistics/vstat_C0242379_<CountryCode><AgeGroup><Gender><Ethnicity><Year>, where the year is included to prevent merging of triples from different years. City and region URIs do not include the year, so each city or region appears exactly once in the graph. Population URIs do include the year, because a city's population can change over time. The current version has population data only for 2024 (EEA) and 2023 (OECD), but the URI structure is designed to accommodate additional years in the future.

A central rule established during the alignment is the no-redefinition rule: entities already present in the knowledge graph must not be redefined in the new TTL files. In RDF, if the same URI is defined in multiple files with different properties, all triples are merged when loaded into the triplestore, which can lead to conflicting information. This rule applies to Country entities (171 already exist), CalendarYear entities (1990, 1995 and 1999 to 2022 are already defined), Chemical entities (the five existing pollutants) and Source entities (EEA-2025, OECD-2025). The pipelines reference these by URI but never emit new type or label triples for them.

The corrected relationship structure follows the Lung-CABO ontology. For Vital Statistics, the sio:SIO_000300 (*has_value*) relationship links from the Disease entity to each Vital Statistics instance. Each instance then links to its CalendarYear, People entity and Country. Incidence and mortality rates are stored as data properties directly on the instance. For Chemical-Location Associations, each CLA instance links to its Chemical, its geographic entity, its Source, its CalendarYear and its Unit, with the concentration value stored using sem-lucia:value.

City deduplication was applied to the EEA data. Several types of duplicates were identified: cities with trailing semicolons, accent variations and inconsistent Unicode representations of Turkish and other special characters. A normalisation step strips trailing punctuation, normalises Unicode characters to their canonical forms and applies consistent transliteration rules. After deduplication, the EEA data contains 3,143 unique Geopolitical Region entities across 30 countries. All pipeline scripts referenced in this chapter are available in the LUCIA project repository at https://medal.ctb.upm.es/internal/gitlab/lucia/domenic_schmidt under data_curation/scripts/. Subsequent footnotes reference scripts by their relative path within this directory.

3.3 Data Pipelines

All four pipelines follow the same two-stage architecture. The first stage transforms raw source data into a clean, standardised CSV. The second stage converts the CSV into RDF triples following the ontological rules from Section 3.2. Shared functionality is centralised in utility modules that provide URI builder functions, UMLS identifier lookups and a registry of existing entities that must not be redefined. This section describes each pipeline in turn.

3.3.1 EEA Air Quality Pipeline

The European Environment Agency publishes air quality monitoring data as Parquet files through a public download service. The data covers annual mean concentrations for multiple pollutants across thousands of measurement stations in Europe. For this thesis, data was collected for ten pollutants: PM2.5,

PM10, PM10 with arsenic, PM10 with cadmium, PM10 with nickel, PM10 with lead, benzo[a]pyrene, benzene, nitrogen dioxide and ozone.

The collection is performed in two steps. First, the EEA data catalogue is queried² to identify all available download URLs for the selected pollutants and years. Second, the Parquet files are downloaded³ with support for resumption, so that interrupted transfers can be continued without re-downloading completed files. A minimum-year filter restricts the collection to 2013 to 2024, the range where consistent annual mean data is available across most countries.

The downloaded Parquet files are processed into a single consolidated CSV⁴. Station-level measurements are aggregated to city-level annual means, since the knowledge graph operates at the city level rather than the station level. City names are normalised using the deduplication rules from Section 3.2. Population data for each city is obtained from Eurostat's Local Administrative Unit (LAU) dataset for 2024, achieving 89.7% coverage across the 3,143 unique cities. The output CSV contains one row per city-year-pollutant combination, with columns for country code, city name, chemical identifier, year, concentration value, unit of measurement and population.

The CSV is converted into RDF triples⁵ following the ontological rules from Section 3.2. For each row, one ChemicalLocationAssociation instance is created, linking to the corresponding Chemical (via `sio:SIO_000628`), Geopolitical Region (via `sio:SIO_000628`), CalendarYear (via `sio:SIO_000679`), Unit (via `sio:SIO_000008`) and Source (via `sio:SIO_000253`). The concentration value is stored as a decimal literal using the `sem-lucia:value` property. Geopolitical Region entities are created once per unique city. Each region links to its parent Country via `sio:SIO_000061` and to a Population instance via `sio:SIO_000216`. The Population instance stores the 2024 population as a numeric value and links to CalendarYear 2024 via `sio:SIO_000679`.

The final EEA TTL file contains 117,609 CLA instances, 3,143 Geopolitical Region entities and 2,791 Population instances across 30 European countries, covering 2013 to 2024. The file is approximately 96 MB. The difference between 3,143 regions and 2,791 Population instances is due to 352 cities for which no matching LAU population data could be found. These cities still have CLA instances but lack a population value.

3.3.2 ECIS Cancer Statistics Pipeline

The European Cancer Information System provides cancer incidence and mortality data from national and regional cancer registries across Europe, maintained by the European Commission's Joint Research Centre.

Data collection was performed in two phases. In the first phase, the 2024 estimates were downloaded through the ECIS data explorer, providing country-level incidence and mortality rates grouped by six age bands (0-14, 15-29, 30-44, 45-59, 60-74, 75-85+) and gender, covering 30 European countries. In the second phase, historical data was added from the historical data explorer, which provides data at both country-level and individual registry-level. The historical data uses more detailed age groupings for the older population (75-89, 90-95+)

² See `eea/01_collect_urls.py`.

³ See `eea/02_download_parquets.py`.

⁴ See `eea/eea_pipeline.py`.

⁵ See `eea/eea_to_ttl.py`.

than the 2024 estimates (75-85+). To maintain consistency with the existing ontology, the historical age groups were mapped to the coarser grouping, consolidating records for ages 75-95+ under the 75-85+ bracket. Historical records extend as far back as 1953 for some registries, though most begin reporting in the 1980s or 1990s. Country-level records were mapped to Countries (ncit:C25464) and registry-level records to Geographic Regions (sio:SIO_000414). In total, the historical data added 106 registry entities across 35 countries, including countries like Ukraine, Bosnia-Herzegovina and Liechtenstein that are not present in the 2024 estimates.

The 2024 estimates and historical records are merged into a single CSV⁶. Gender values are normalised, registry names are cleaned and entries with missing values are excluded. Each row represents one Vital Statistics observation for a specific combination of disease (malignant neoplasm of lung, C0242379), country or registry, age group, gender, ethnicity and year. Ethnicity is set to 'undefined' for all ECIS records, since European cancer registries do not collect ethnicity data.

Each row is converted into a Vital Statistics instance⁷. The Disease entity points to the Vital Statistics instance via sio:SIO_000300. Each instance links to its CalendarYear (via sio:SIO_000679), People entity (via sio:SIO_000229) and Country or Geographic Region (via sio:SIO_000061). Incidence and mortality rates are stored as xsd:float data properties using sem-lucia:incidence and sem-lucia:mortalityrate. People entities are created once per unique combination of age group, gender and ethnicity. The URI for each Vital Statistics instance includes the year, so that adding data for a new year does not cause RDF to merge the new triples with existing instances.

The final ECIS TTL file contains 32,304 Vital Statistics instances: 9,228 national-level records linked to Country entities and 23,076 registry-level records linked to Geographic Region entities, covering 35 countries, 106 registries and a temporal range from 1953 to 2024. The data produces 12 People entities (6 age groups times 2 genders, all with ethnicity 'undefined'). The file is approximately 21 MB.

3.3.3 OECD Expose Pipeline

The OECD publishes population-level exposure estimates for air pollutants, combining monitoring data with atmospheric modelling to estimate concentrations at the NUTS2/TL2 regional level. The data is published through the OECD.Stat portal and covers OECD member countries and selected partner economies. The dataset provides two types of measurements for each region and year: the mean annual concentration in micrograms per cubic metre and the percentage of the population exposed above a given threshold. For PM2.5, both measurement types are available from 1990 onwards. For PM10 and NO2, data is available from 2021 onwards.

The main processing challenge is mapping OECD-specific TL2 region codes⁸ to human-readable region names using the OECD regional metadata. Population data for each region was sourced from Eurostat's NUTS2 2023 dataset, which covers EU member states and candidate countries. For non-EU OECD members (e.g. Australia, Canada, Japan, South Korea), Eurostat does not provide regional

⁶ See `ecis/ecis_pipeline.py`.

⁷ See `ecis/ecis_to_ttl.py`.

⁸ See `oecd/oecd_exposure_pipeline.py`.

population data. Integrating the OECD Regional Demography TL2 dataset to fill this gap was identified during the project but remains outstanding. The current version therefore has population values for EU regions only.

The TTL generation follows the same structure⁹ as the EEA pipeline. For each row in the CSV, one CLA instance is created, linking to its Chemical, Geographic Region, CalendarYear, Unit and Source (OECD-2025) via the same SIO properties. Geographic Regions are typed as `sio:SIO_000414`, distinct from the EEA's Geopolitical Regions (`sio:SIO_000415`), reflecting that OECD regions are statistical aggregations rather than individual cities. An important difference from the EEA pipeline is the handling of units. The original knowledge graph had all values from both units mixed into a single entity per country. The restructured pipeline creates one separate CLA instance per country-region-chemical-year-unit combination, fully resolving this ambiguity. For Spain alone, a single entity with 28 mixed values is now 28 individual instances, each with a single value clearly linked to its year, unit and region.

The final OECD TTL file contains 14,001 CLA instances, 444 Geographic Region entities and 439 Population instances across more than 40 countries, covering 1990 to 2023 for PM2.5 and 2021 to 2023 for PM10 and NO2. The file is approximately 13 MB. The five missing Population instances correspond to regions in non-EU countries where Eurostat data is not available.

3.3.4 CDC Cancer Statistics Pipeline

The Centers for Disease Control and Prevention provides cancer incidence data for the United States and Puerto Rico through its WONDER (Wide-ranging Online Data for Epidemiologic Research) platform. The data covers lung cancer incidence broken down by five-year age bands (20-24, 25-29, up to 80-84 and 85+), gender and ethnicity (Hispanic, Non-Hispanic). Unlike the ECIS data, the CDC data includes an ethnicity dimension but does not provide mortality rates.

The original data stored all incidence values from 1999 to 2021 in a single Vital Statistics entity per demographic group. A single entity could contain 23 separate incidence values with no way to associate each value with its corresponding year. The pipeline splits each such entity¹⁰ into individual observations, one per year. Each is converted into a Vital Statistics instance following the same relationship pattern as ECIS: the Disease entity points to the instance via `sio:SIO_000300`, and the instance links to its CalendarYear (via `sio:SIO_000679`), People entity (via `sio:SIO_000229`) and the Country entity `US_PRI` (via `sio:SIO_000061`). The incidence value is stored using `sem-lucia:incidence`. Since no mortality data is available, `sem-lucia:mortalityrate` is omitted.

People entities are created for each unique combination of age group, gender and ethnicity. The CDC data produces 54 People entities, considerably more than the 12 from ECIS, because of the finer age groupings and the additional ethnicity dimension. The two sources share the same People entity type (`ncit:C95553`) but their instances differ.

The final CDC TTL file contains 1,184 Vital Statistics instances and 54 People entities, covering 1999 to 2021. The file is approximately 719 KB, making it the smallest of the four source files. The relatively low number compared to ECIS

⁹ See `oecd/oecd_exposure_to_ttl.py`.

¹⁰ See `cdc/cdc_to_ttl.py`.

(1,184 vs 32,304) reflects two factors: the CDC data covers only a single country entity (US_PRI) rather than 35 countries and 106 registries, and some demographic groups have suppressed values due to low case counts, which is standard CDC practice to protect patient privacy.

3.4 Shared Entities and Cross-Source Integration

Several entity types are referenced across multiple data sources. CalendarYear entities appear in all four pipelines, Chemical entities are shared between EEA and OECD, and Country entities are used by EEA, ECIS and OECD. To avoid duplication and enforce the no-redefinition rule, all shared entity definitions are centralised in a single shared TTL file¹¹ (graph_shared.ttl).

The LUCIA knowledge graph already contained CalendarYear instances for 1990, 1995 and 1999 to 2022. Only two additional years had to be created: 2023 and 2024. All four pipelines reference CalendarYear entities by URI without redefining them. The EEA data spans 2013 to 2024, the OECD data 1990 to 2023, the ECIS data 1953 to 2024 and the CDC data 1999 to 2021. The union of these ranges is fully covered by the existing instances plus the two new ones.

Of the ten chemicals used across the pipelines, five already existed in the ontology and five PM10 variants had to be created. Of the countries referenced, 171 already existed and two had to be added (South Korea and Kosovo). Two Source entities (EEA-2025, OECD-2025) are referenced but not redefined.

The complete shared TTL file defines exactly 9 new entities: 2 CalendarYear instances, 5 PM10 chemical variants and 2 Country instances.

3.5 Validation and Summary Statistics

After all four pipelines were implemented, the generated TTL files went through a multi-step validation process combining automated checks with manual review by the domain expert.

The first step was syntactic validation: each TTL file was parsed using a standard RDF library to verify that the generated triples are well-formed. The second step was entity count verification: the domain expert independently counted the expected number of entities based on the source CSVs and compared them against the actual counts in the TTL files. Several discrepancies were caught during early iterations. The initial EEA output contained 31,142 Geopolitical Region entities because the year had been included in the Region URI, creating one region per city-year combination rather than one per city. After correcting the URI pattern, the count matched the expected 3,143. A similar issue occurred in the OECD data, where the initial output contained 11,337 Geographic Region entities instead of the expected 444. The third step was relationship verification: sample entities were inspected in the Virtuoso triplestore interface to verify that relationship patterns matched the corrected ontology. The fourth step was consistency checking across sources, verifying that URIs for shared entities match exactly across all files. Table 3.1 summarises the output of all four pipelines. The Total of 165,098 refers to the primary instances created, the Chemical-Location Associations and Vital

¹¹ See shared/generate_shared_ttl.py.

Statistics; Region, Population, People and shared entities are listed per source but not counted in that total.

Table 3.1: Summary of pipeline outputs

Source	Entity Type	Instances	Countries	Temporal Range	Pollutants	File Size
EEA	ChemicalLocationAssociation	117,609	30	2013-2024	10	96 MB
	Geopolitical Region	3,143				
	Population	2,791		2024 only		
ECIS	Vital Statistics	32,304	35	1953-2024	-	21 MB
	Geographic Region (registry)	106				
	People	12				
OECD	ChemicalLocationAssociation	14,001	40+	1990-2023	3	13 MB
	Geographic Region	444				
	Population	439		2023 only		
CDC	Vital Statistics	1,184	1	1999-2021	-	719 KB
	People	54				
Shared	CalendarYear / Chemical / Country	9	2	2023-2024	5	2 KB
Total	All entity types	165,098		1953-2024		~131 MB

In total, the four pipelines produced 165,098 new entity instances. Of these, 131,610 are Chemical-Location Associations (117,609 from EEA and 14,001 from OECD) and 33,488 are Vital Statistics (32,304 from ECIS and 1,184 from CDC). The combined output is approximately 131 MB of TTL data.

4 GNN-Based Link Prediction

This chapter describes the methodology used to train and evaluate Graph Neural Network models for link prediction on the LUCIA knowledge graph. The chapter covers six aspects of the approach. Section 4.1 formulates the link prediction task in formal terms. Section 4.2 describes how the curated TTL files from Chapter 3 are parsed and combined with the biological subgraph. Section 4.3 explains how the resulting graph is converted into a format suitable for GNN training, including the construction of reification nodes and node features. Section 4.4 presents the three model architectures used in this thesis in detail: Dot Product, TransE and R-GCN. Chapter 2 introduced these models conceptually. This chapter describes how they are implemented and configured for the LUCIA knowledge graph. Section 4.5 details the training procedure. Section 4.6 specifies the evaluation protocol.

4.1 Problem Formulation

The task addressed in this thesis is link prediction on a heterogeneous knowledge graph. Given a set of known triples, the goal is to predict which additional triples are likely to be true but have not been recorded.

Formally, let $G = (E, R, T)$ denote a knowledge graph, where E is the set of entities, R is the set of relation types and T is the set of observed triples. Each triple $t = (h, r, t)$ consists of a head entity h , a relation type r and a tail entity t . A link prediction model learns a scoring function $f: E \times R \times E \rightarrow \mathbb{R}$ that assigns a real-valued score to any candidate triple. Higher scores indicate higher plausibility.

At inference time, the model is used to rank candidate triples. For each test triple (h, r, t) , the model is queried with the correct head and relation and asked to score every possible tail entity. The rank of the correct tail in this ranking serves as the basis for evaluation.

The LUCIA knowledge graph contains entities of multiple types (genes, diseases, chemicals, regions, etc.) and relations of multiple types (*has_association*, *refers_to*, *detected_finding*, *in_pathway* etc.). This makes it a heterogeneous, multi-relational graph. The link prediction task on such a graph is more complex than on a homogeneous graph, because the model must learn to distinguish between different relation types and respect the type constraints of each relation. For example, the relation *associated_with* only connects Gene entities to Disease entities, while *exposure_in* connects Chemical entities to Geographic or Geopolitical Regions.

The central question this thesis addresses is whether the inclusion of environmental data improves predictions of gene-disease associations and what novel links the trained models predict. To answer this, three models with increasing capacity are compared on the same graph: a minimum-capacity Dot Product baseline, a translation-based TransE model and a relation-aware R-GCN model. An ablation study then trains R-GCN on different subsets of the graph (combined, biological-only, environmental-only) to isolate the contribution of the environmental data. Beyond evaluation, the best-performing model is used to generate novel link predictions across both biological and environmental relations.

4.2 Data Preparation

The first step in the GNN pipeline is to convert the curated TTL files from Chapter 3 into a unified node and edge representation. This is implemented in a dedicated parsing script¹². The script handles three things: parsing the environmental TTL files, parsing the biological SPARQL exports and merging the two subgraphs into a single graph.

4.2.1 Parsing the Environmental Subgraph

The five environmental TTL files (`graph_shared.ttl`, `graph_CDC.ttl`, `graph_ECIS.ttl`, `graph_OECD.ttl`, `graph_EEA.ttl`) are loaded into a single rdflib graph. After parsing, the combined graph contains 1,451,650 triples.

From this RDF graph, the script extracts nodes and edges into a tabular representation suitable for downstream processing. Nodes are identified by their URI and assigned a type. The type is derived from two sources. If the entity has an explicit *rdf:type* triple in the data, the type is read directly from that triple. If no explicit type is present, the type is inferred from the URI structure. For example, URIs containing the path segment *#country/gpr/* are mapped to the type *GeopoliticalRegion* and URIs containing *#calendaryear/* are mapped to *CalendarYear*. This inference resolves the type for 124 entities that lack explicit type triples in the source data.

Edges are extracted from all triples whose object is a URI rather than a literal. Triples where the predicate is *rdf:type* or *rdfs:label* are skipped, since these are used for typing and labelling, rather than for representing relationships. Triples with literal objects (such as exposure values, incidence rates and category labels) are stored as edge attributes on the corresponding entities, not as edges.

After this extraction, the environmental subgraph contains 172,183 nodes and 538,897 edges.

4.2.2 Parsing the Biological Subgraph

The biological subgraph is exported from the LUCIA SPARQL endpoint as a set of seven TTL files. Each file covers one type of biological relationship: gene-disease associations, variant-disease associations, disease-gene-pathway connections, pathway-disease links, biomarker-disease associations, disease-chromosomal rearrangements and disease-gene fusions.

These files use the SPARQL Results format, rather than plain RDF. Each result row is parsed and converted into a typed edge. For example, the `gene_disease_assoc.ttl` file produces 17,701 unique Gene-Disease edges, each carrying an *associated_with* relationship.

The biological subgraph contains 18,759 nodes and 80,014 edges. The largest source is Gene-Pathway memberships with 54,005 edges, followed by Gene-Disease associations with 17,701 edges and Disease-GeneFusion links with 4,289 edges. Disease-Chromosomal Rearrangement contributes 2,309 edges, Variant-Disease and Variant-Gene associations contribute 1,516 edges, GeneProduct-Pathway and Pathway-Disease contribute 170 edges and Biomarker-Disease associations contribute 24 edges.

¹² See `01_parse_kg.py`.

4.2.3 Merging the Two Subgraphs

The biological and environmental subgraphs share Disease entities. A specific lung cancer subtype, like Non-Small Cell Lung Carcinoma, may appear in both subgraphs with the same UMLS identifier, but slightly different URIs. To prevent duplicate Disease entities in the merged graph, the script identifies Disease entities by their UMLS code and unifies them. During the merge, biological edges that reference a Disease are remapped to the corresponding environmental Disease URI where one exists. This process bridges one Disease entity across the two subgraphs.

4.2.4 Adding Subtype Bridge Edges

A structural problem becomes apparent at this point. The environmental subgraph references only a single Disease entity, the parent term Malignant neoplasm of lung (UMLS C0242379). All cancer incidence and mortality statistics from ECIS and CDC are connected to this single entity. The biological subgraph, in contrast, references 42 specific lung cancer subtypes (Non-Small Cell Lung Carcinoma, Adenocarcinoma of lung, Small cell carcinoma of lung, etc.). Without explicit links between the parent term and the subtypes, the GNN cannot propagate environmental information to the biological side of the graph through Disease nodes.

To resolve this, the script adds 35 explicit *subtype_of* edges. Each edge connects a specific lung cancer subtype to the parent Disease entity. Of the 42 subtype-level disease entities, 35 are strict malignant subtypes of lung cancer and therefore receive a *subtype_of* edge to the parent. The remaining seven are not strict subtypes: they are pre-cancerous conditions (hyperplasia, metaplasia), benign tumours (hamartoma of lung, pulmonary sclerosing hemangioma) and related non-carcinoma entities (pulmonary lymphangioleiomyomatosis, Proteus syndrome). These remain in the graph as Disease nodes but are deliberately left out of the subtype hierarchy, which is why the number of bridge edges is 35 rather than 42. These edges turn the parent Disease into a hub that bridges the two subgraphs. During GNN message passing, environmental signals reaching the parent Disease can then propagate to the biological subtypes through these bridge edges. This is a small but critical structural addition. Without it, the two subgraphs would remain effectively disconnected at the Disease layer.

4.2.5 Output

The final, merged graph is written to two CSV files. The nodes.csv file lists all 190,941 entities, with their type and label. The edges.csv file lists all 618,946 edges with source, source type, relation, destination, destination type and any literal attributes. A schema_summary.json file records the count of each node type and edge type, for reference.

This intermediate representation is the input for the next step, where the graph is restructured and converted into a format suitable for GNN training.

4.3 Graph Construction

The merged graph from Section 4.2 cannot be used directly for GNN training. It contains several entity types that are n-ary reifications: nodes whose purpose is to connect multiple other entities and carry literal values. For example, a single ChemicalLocationAssociation entity links a Chemical, a Region, a CalendarYear, a Source and a Unit, with the actual measurement value stored as a literal attribute. Similarly, Gene-Disease Associations in the biological

subgraph carry a numeric association score from DisGeNET that quantifies the strength of evidence for each gene-disease link.

A common approach in prior work is to collapse these reifications into direct edges between the meaningful entities they connect. This simplifies the graph but discards the literal values, because standard R-GCN layers cannot process edge features. The approach taken in this thesis is different. Rather than collapsing the reifications, they are preserved as first-class nodes in the graph. Each reification node carries its literal values as node features, which the R-GCN can access through its feature projection mechanism. This design choice preserves the full granularity of the data at the cost of a larger graph.

This section describes how the reification nodes are handled, how node features are constructed and how the resulting graph is built into the PyTorch Geometric format used for training¹³.

4.3.1 Preserving N-ary Reifications

Four types of reification nodes are present in the graph. Two come from the environmental subgraph and two from the biological subgraph.

ChemicalLocationAssociation (CLA) nodes represent individual air quality measurements. Each CLA links a Chemical to a geographic entity (a Geopolitical Region from EEA or a Geographic Region from OECD) and to a CalendarYear, carrying the measured pollutant concentration as a literal value. The 131,610 CLA nodes remain in the graph. Each receives a 1-dimensional feature vector containing its concentration value.

VitalStatistics nodes represent cancer registry observations. Each links a Disease to a geographic entity (a Country or Geographic Region), a CalendarYear and a People demographic group, carrying incidence and mortality rates as literal values. The 33,488 VitalStatistics nodes remain in the graph. Each receives a 2-dimensional feature vector containing its incidence and mortality rates.

Gene-Disease Association (GDA) nodes are created during graph construction to replace the direct Gene-Disease edges from Section 4.2. In the merged graph, each Gene-Disease edge carries a GDA score from DisGeNET that quantifies the evidence for the association. Since R-GCN cannot use edge features, these edges are split into two-hop paths: Gene to GDA to Disease. Each of the 17,701 GDA nodes receives a 1-dimensional feature vector containing the DisGeNET association score. This score ranges from 0.01 to 0.80, where higher values indicate stronger evidence.

Variant-Disease Association (VDA) nodes follow the same logic. Each replaces a direct Variant-Disease edge with a two-hop path: Variant to VDA to Disease. The 733 VDA nodes each receive a 2-dimensional feature vector containing the Disease Specificity Index (DSI) and Disease Pleiotropy Index (DPI) from DisGeNET. Each VDA node also links to the Gene in which the variant is located, preserving the variant-gene relationship from the source data.

Two node types from the environmental subgraph are removed during graph construction. Source nodes (EEA-2025, OECD-2025) are administrative metadata and carry no information relevant to link prediction. Population nodes

¹³ See 02_build_graph.py.

are dropped as separate entities, but their population values are incorporated as features on the corresponding Region nodes.

4.3.2 Node Features

Ten of the eighteen node types in the final graph carry feature vectors derived from literal values in the source data. The remaining eight node types use learnable embeddings without input features. All feature vectors are normalised using z-score standardisation (subtracting the mean and dividing by the standard deviation across all nodes of the same type) so that each feature dimension has zero mean and unit variance.

Reification nodes carry the features described in Section 4.3.1: CLA nodes have 1-dimensional concentration features, VitalStatistics nodes have 2-dimensional incidence and mortality features, GDA nodes have 1-dimensional association score features and VDA nodes have 2-dimensional DSI and DPI features.

People nodes receive a 2-dimensional feature vector. The first dimension encodes the midpoint of the age group, normalised to $[0, 1]$. For example, the 60-74 age group receives a value of 0.67. The second dimension encodes gender as a binary value (0 for male, 1 for female). There are 66 People nodes in total, covering 12 ECIS demographic groups and 54 CDC demographic groups.

Variant nodes receive a 3-dimensional feature vector. The first dimension encodes the chromosome number, normalised to $[0, 1]$ (chromosome 1 maps to $1/24$, chromosome X to $23/24$). The second dimension encodes the consequence type of the variant (missense, intron, frameshift, etc.) as a normalised categorical value. The third dimension encodes the chromosomal start position, normalised by the maximum chromosome length.

ChromoRearr nodes receive a 1-dimensional feature vector encoding the rearrangement type (translocation, inversion, derivative chromosome or deletion) as a normalised categorical value.

Region nodes (both Geopolitical Regions and Geographic Regions) receive a 1-dimensional feature vector containing the population of the region. Population data is sourced from Eurostat LAU 2024 for EEA cities (89.7% coverage) and Eurostat NUTS2 2023 for OECD regions.

CalendarYear nodes receive a 1-dimensional feature vector. The single feature is the year value normalised as $(\text{year} - 1950) / 80$, giving the model an explicit temporal signal.

The remaining node types (Gene, Disease, Country, Chemical, GeneFusion, Pathway, GeneProduct and Biomarker) have no input features. Their representations are learned from scratch as 128-dimensional embeddings during training.

4.3.3 Building the PyTorch Geometric HeteroData

The final graph is constructed as a HeteroData object from PyTorch Geometric. Each node type is stored as a separate index space and each edge type is stored as a typed edge index. This native heterogeneous representation is necessary for the R-GCN model, which treats different relation types differently during message passing.

For each node type, the script assigns a contiguous integer index to every entity. The labels and original URIs are kept in a separate JSON file¹⁴ so that predictions can be traced back to readable entities. For node types with feature vectors, the feature matrix is attached as `data[node_type].x`. For node types without features, only the count is stored and PyG's `RGCNConv` handles them with learnable embeddings.

For each of the 24 edge types, the source and destination indices are stored as a $(2, \text{num_edges})$ tensor. The edge types are organised as triples of $(\text{source_type}, \text{relation}, \text{destination_type})$, following PyG's convention for heterogeneous graphs.

4.3.4 Final Graph Statistics

The final graph contains 206,151 nodes across 18 node types and 631,659 edges across 24 edge types. The breakdown by node type is shown in Table 4.1. Figure 4.1 shows the overall schema of the graph, organised into biological, bridge and environmental layers.

Table 4.1: Node types in the Lung-CABO knowledge graph

Node Type	Layer	Count	Features	Description
ChemicalLocationAssoc.	Environmental	131,610	1-dim (concentration)	Air quality measurements
VitalStatistics	Environmental	33,488	2-dim (inc., mortality)	Cancer registry observations
GeneDiseaseAssociation	Biological	17,701	1-dim (GDA score)	Gene-disease evidence scores
Gene	Biological	10,099	learnable	Human genes
GeneFusion	Biological	4,266	learnable	Gene fusion events
GeoPoliticalRegion	Environmental	3,143	1-dim (population)	Cities (EEA)
ChromoRearr	Biological	1,993	1-dim (type)	Chromosomal rearrangements
Pathway	Biological	1,492	learnable	Biological pathways
VariantDiseaseAssoc.	Biological	733	2-dim (DSI, DPI)	Variant-disease evidence
Variant	Biological	709	3-dim (chrom, cons, pos)	Sequence variants
GeographicRegion	Environmental	518	1-dim (population)	Statistical regions (OECD)

¹⁴ See `node_id_maps.json`.

GeneProduct	Biological	133	learnable	Gene products / proteins
CalendarYear	Temporal	72	1-dim (year)	Temporal references
People	Demographic	66	2-dim (age, gender)	Demographic groups
Country	Environmental	51	learnable	National-level entities
Disease	Bridge	43	learnable	Lung cancer subtypes
Biomarker	Biological	24	learnable	Disease biomarkers
Chemical	Environmental	10	learnable	Air pollutants
Total	18 node types	206,151		

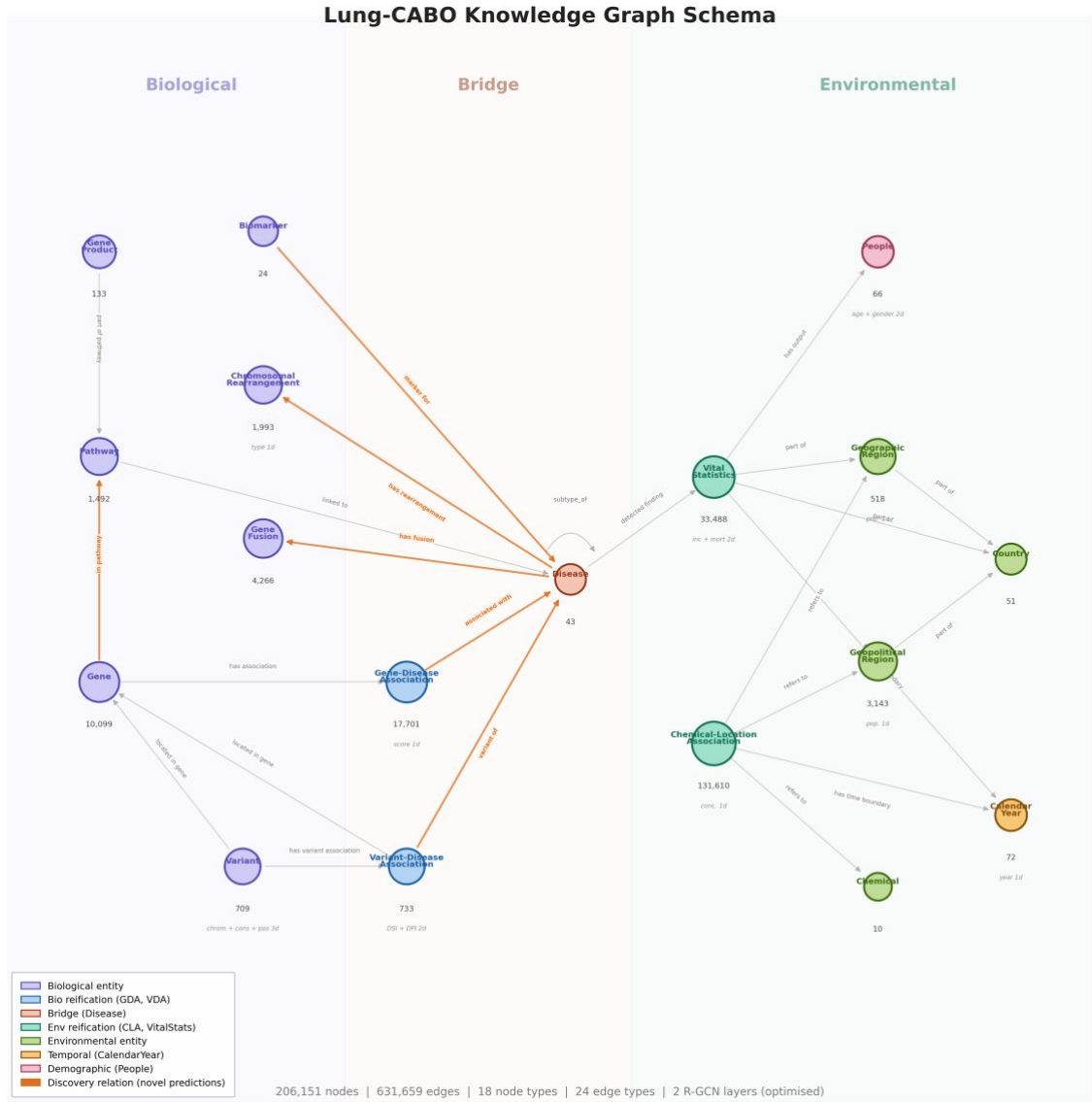


Figure 4.1: Schema of the Lung-CABO knowledge graph

The largest edge types are CLA-Chemical and CLA-CalendarYear (131,610 each), CLA-GeoPoliticalRegion (117,609), Gene-Pathway (54,005) and Disease-VitalStatistics (33,488). The smallest are Pathway-Disease (2 edges), Biomarker-Disease (24 edges) and Disease-subtype_of-Disease (35 edges). This imbalance reflects the underlying data sources. The environmental subgraph contributes the majority of edges through the CLA and VitalStatistics reification nodes, while some biological relations come from small lookup tables.

An important structural property of the graph is the Disease bridge. The 35 subtype_of edges connect specific lung cancer subtypes to the parent Disease entity. On the biological side, Gene-Disease associations reach the subtypes through GDA nodes. On the environmental side, VitalStatistics nodes connect the parent Disease to geographic regions. This makes the parent Disease a high-degree hub through which environmental signals can, in principle, propagate to the biological subtypes during R-GCN message passing.

This graph is the input to all three models described in Section 4.4. It is saved as a single PyTorch file¹⁵ along with the node ID mappings, ready to be loaded for training.

4.4 Model Architectures

This section describes the three models compared in this thesis. All three operate on the same graph and produce a scoring function $f(h, r, t)$ that ranks candidate triples. All three operate on the same graph and produce a scoring function $f(h, r, t)$ that ranks candidate triples. The three models differ in how they encode the graph structure and how they incorporate relation information. Dot Product is the minimum-capacity baseline that uses only entity embeddings. TransE adds a translation-based relation model. R-GCN adds relation-specific message passing on top of that.

All three models embed entities into a 128-dimensional vector space. The choice of dimension follows common practice in the knowledge graph embedding literature.

4.4.1 Dot Product

The Dot Product model is the simplest of the three. Each entity e is represented by a single embedding vector e_emb in $\mathbb{R}^{128}$. Relations are not modelled at all. The score for a triple (h, r, t) is the inner product of the head and tail embeddings:

$$f(h, r, t) = \text{sum}(h_emb * t_emb)$$

Where $*$ denotes element-wise multiplication. The relation r is ignored entirely. A triple is scored as plausible whenever the head and tail embeddings are similar in the learned vector space.

Two design choices follow from this. First, the model is permutation-invariant in head and tail. The score for (h, r, t) is the same as the score for (t, r, h) . The model cannot distinguish the direction of a relationship. Second, the model assigns the same score to (h, r_1, t) and (h, r_2, t) for any two relations r_1 and r_2 . The relation type carries no information.

These limitations are intentional. Dot Product serves as a sanity check. If a more expressive model fails to outperform it, the additional complexity adds no real predictive power. The implementation initialises entity embeddings with Xavier uniform initialisation, which samples initial values from a uniform distribution scaled by the input and output dimensions to keep gradients stable during early training. No relation embeddings are used.

4.4.2 TransE

TransE represents each relation as a translation in embedding space. In addition to the entity embeddings, the model maintains a relation embedding r_emb in $\mathbb{R}^{128}$ for each relation type. The score is the negative L2 distance between the translated head and the tail:

$$f(h, r, t) = -||h_emb + r_emb - t_emb||_2$$

¹⁵ See `hetero_graph.pt`.

When the triple is true, the head plus the relation should land close to the tail in embedding space. The negative distance turns this into a score where higher means more plausible. TransE is one of the earliest knowledge graph embedding methods and remains a strong baseline despite its simplicity.

The model has known limitations. It cannot represent symmetric relations cleanly, because that would require the relation embedding to be the zero vector, which removes the relation information entirely. It also has difficulty with one-to-many relations, where a single head can be connected to many different tails through the same relation. Despite these limitations, TransE often performs well on biomedical knowledge graphs because many relations of interest are asymmetric and many-to-many.

The TransE implementation also uses Xavier uniform initialisation, for both entity and relation embeddings. Training uses a margin-based ranking loss, described in Section 4.5.

4.4.3 R-GCN with DistMult Decoder

The R-GCN model is the primary model of this thesis and structurally different from the two baselines. It is the most expressive of the three architectures and the one from which the best results are expected. It uses an encoder-decoder architecture. The encoder is a Relational Graph Convolutional Network that produces refined node embeddings by aggregating information from each node's neighbourhood. The decoder is a DistMult scoring function that converts the encoded embeddings into triple scores.

The encoder operates as follows. Each node in the graph is initially represented by a 128-dimensional vector. For nodes with input features (CLA, VitalStatistics, GDA, VDA, People, Variant, ChromoRearr, Regions and CalendarYears), the initial vector is the sum of two components: a learnable embedding from a lookup table and a linear projection of the input features into the 128-dimensional space. For nodes without features (Genes, Diseases, Countries, Chemicals, etc.), the initial vector consists only of the learnable embedding. This combination allows the model to use the quantitative information already encoded in features (such as actual pollutant concentrations or GDA scores) while still learning task-specific corrections.

The architecture of the R-GCN encoder was determined through a hyperparameter search using Bayesian optimisation (described in Section 4.5.5). The search evaluated configurations with 2 to 5 layers and identified 2 layers as optimal. The encoder therefore applies two layers of relational graph convolution. In each layer, every node aggregates messages from its neighbours, grouped by relation type. For each of the 24 relation types in the graph, a separate transformation matrix W_r is applied to the messages from neighbours connected through that relation. The aggregated message is added to a self-connection term, passed through a ReLU (Rectified Linear Unit) activation function, which sets all negative values to zero, and followed by dropout at a rate of 0.15.

To control the number of parameters, the relation-specific weight matrices use basis decomposition with 6 bases. Each W_r is expressed as a linear combination of 6 shared basis matrices, with relation-specific coefficients. This reduces the parameter count from 24 separate 128 by 128 matrices to 6 basis matrices plus 24 sets of 6 coefficients. Basis decomposition is the regularisation strategy proposed in the original R-GCN paper [28]. It controls overfitting on

relations with few training examples, which is relevant here because some relation types have only dozens of edges.

After two layers of message passing, each node embedding incorporates information from its 2-hop neighbourhood in the graph. This is sufficient to cover the key biological paths (Gene to GDA to Disease), though the full cross-domain path from Chemical through CLA, Region, VitalStatistics to Disease spans four hops and exceeds the 2-hop receptive field. The implications of this limited receptive field are discussed in Chapter 5.

The decoder is a DistMult scoring function:

$$f(h, r, t) = \text{sum}(z_h * r_emb * z_t)$$

Where z_h and z_t are the encoded node embeddings from the R-GCN encoder and r_emb is a learnable relation embedding. This decoder is the canonical choice from the original R-GCN paper [28]. It is simple, efficient and integrates naturally with the bilinear interaction patterns produced by the encoder.

The three models are designed as a progression. Dot Product uses no relation information at all. TransE adds a translation-based relation model on shallow entity embeddings. R-GCN adds a bilinear relation model (DistMult) on entity embeddings that have been refined by message passing through the graph. Each model adds one component to its predecessor. This makes the three-way comparison a direct test of which components contribute to predictive performance.

The three models are summarised in Table 4.2.

Table 4.2: Summary of the three model architectures

Model	Encoder	Decoder	Score Function	Relation Model
Dot Product	None (lookup embedding)	Inner product	$f(h,r,t) = \text{sum}(h * t)$	None
TransE	None (lookup embedding)	Translation dist.	$f(h,r,t) = - h + r - t $	Translation vector
R-GCN	2-layer R-GCN, basis decomp. (6 bases)	DistMult	$f(h,r,t) = \text{sum}(z_h * r * z_t)$	Diagonal bilinear

4.5 Training Procedure

This section describes how the three models are trained. The training procedure is similar across the three models, with model-specific differences in the loss function. All models share the same data splits, optimiser, batch size, learning rate and number of epochs.

4.5.1 Data Splits

The 631,659 triples in the graph are split randomly into training (80%), validation (10%) and test (10%) sets. This produces 505,329 training triples, 63,165 validation triples and 63,165 test triples. The split is done with a fixed random seed for reproducibility.

The split is done at the triple level rather than the entity level. This means a Gene entity may appear in both training and test triples, but the specific (Gene, *associated_with*, Disease) triples in the test set are not seen during training. This is the standard transductive setting in knowledge graph link prediction. It tests whether the model can predict missing edges in a known graph, not whether it can generalise to unseen entities.

4.5.2 Negative Sampling

Link prediction is trained with a contrastive objective. The model learns to assign higher scores to true triples than to corrupted ones. For each positive triple (h, r, t) in a training batch, negative triples are generated by corrupting either the head or the tail entity. The choice of which side to corrupt is random with equal probability.

The negative sampling in this thesis uses a type-aware strategy. When corrupting the tail of a triple, the replacement entity is sampled uniformly from the set of entities that have the correct type for that relation. For example, when corrupting the tail of a (Gene, *has_association*, GDA) triple, the replacement is drawn only from the 17,701 GDA nodes, not from all 206,151 entities in the graph. This prevents the model from being rewarded for trivially rejecting type mismatches. Without type-aware sampling, the model could achieve high scores simply by learning that a Gene cannot be connected to a CalendarYear, which is not a useful prediction capability.

The same type-aware constraint is applied consistently across all three models and during evaluation, so that the comparison between models remains fair.

A subtle issue with random corruption is that some generated negatives may accidentally be true triples. This is a known limitation of random negative sampling. In practice, the proportion of accidental positives is small relative to the total number of entities within each type and the model is trained to handle this noise.

4.5.3 Loss Functions

The three models use two different loss functions, chosen to match each model's scoring properties.

TransE uses a margin-based ranking loss. For each batch, the loss is the average of $\max(0, \text{margin} - \text{pos_score} + \text{neg_score})$ across all positive-negative pairs, with margin set to 1.0. The loss penalises cases where the negative triple receives a higher score than the positive triple by less than the margin. This pushes the scores of true and false triples apart, which is the natural objective for a distance-based scoring function. Figure 4.2 shows the TransE training loss over 200 epochs.

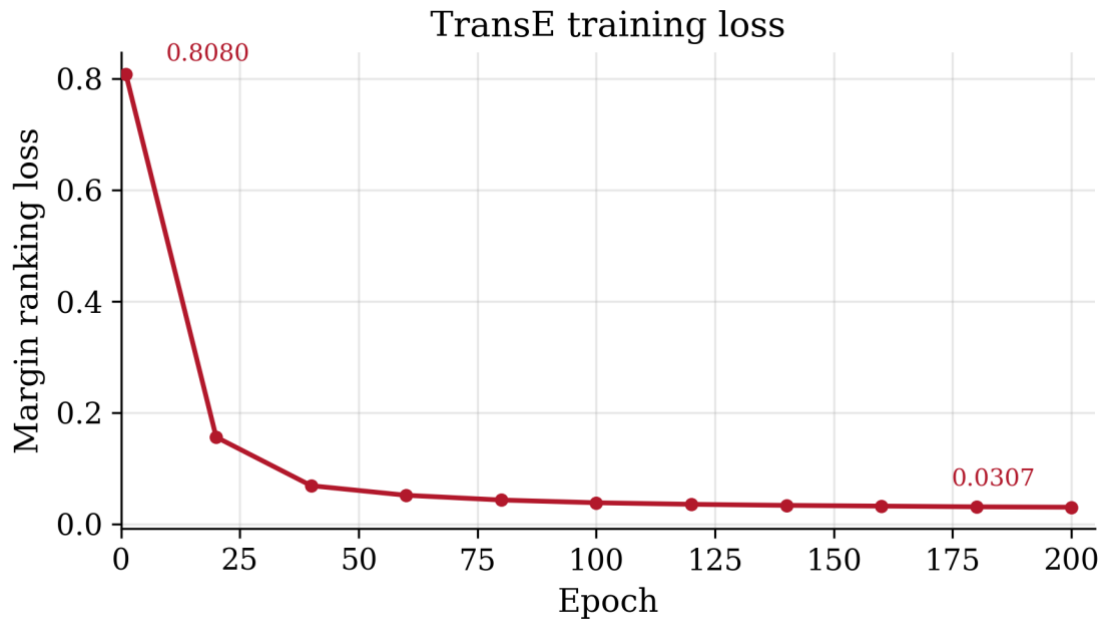


Figure 4.2: TransE training loss

Dot Product and R-GCN both use a binary cross-entropy loss with sigmoid scoring. The positive score is passed through a sigmoid to produce a probability and the negative log-likelihood is minimised against the target value of 1 for positives and 0 for negatives. This is more appropriate for the bilinear scoring functions used by these two models, which produce real-valued scores rather than distances. Figure 4.3 and Figure 4.4 show the training loss curves for Dot Product and R-GCN over 200 epochs.

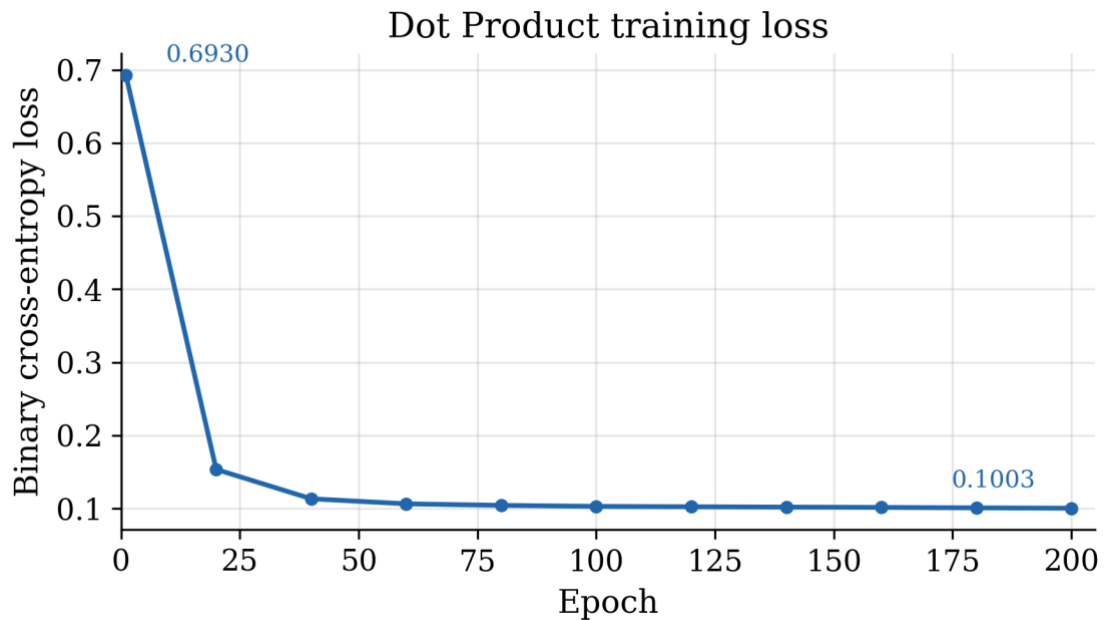


Figure 4.3: Dot Product training loss

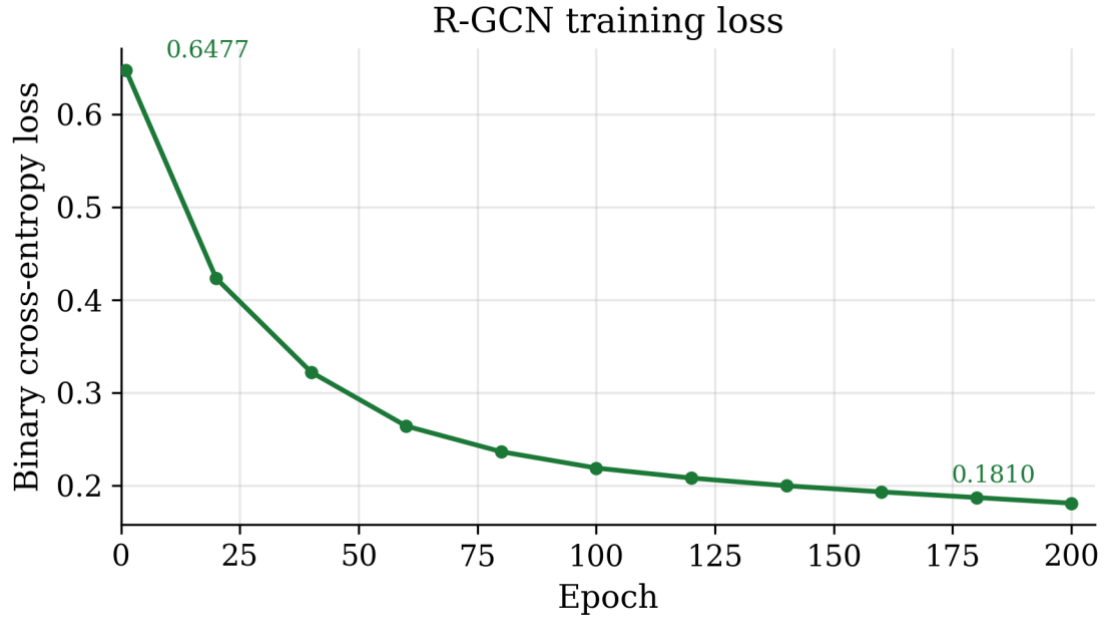


Figure 4.4: R-GCN training loss

The choice of loss function follows common practice in the knowledge graph embedding literature. Margin-based losses pair naturally with translation models like TransE. Cross-entropy losses pair naturally with bilinear scoring functions. Both Dot Product and R-GCN's DistMult decoder fall into this category.

4.5.4 Optimisation

All three models are trained with the Adam optimiser, an adaptive gradient descent method that adjusts the learning rate for each parameter individually based on estimates of the first and second moments of the gradients. The baseline models use a learning rate of 0.001 with 10 negatives per positive. The R-GCN uses a learning rate of 0.0003, with 5 negatives per positive. The batch size is 4,096 positive triples for all models.

For R-GCN, gradient clipping is applied with a maximum norm of 1.0. This is important for relational graph convolutions because the gradients can become unstable on relations with very different connectivity patterns. Clipping prevents single high-gradient steps from disrupting the trained embeddings of low-frequency relations. Dot Product and TransE do not require gradient clipping, because their scoring functions are simpler and produce more stable gradients.

All model weights are initialised with Xavier uniform initialisation. No learning rate schedule is used.

4.5.5 Hyperparameters

The hyperparameters are summarised in Table 4.3.

Table 4.3: Hyperparameters

Hyperparameter	Dot Product	TransE	R-GCN
Architecture			

Embedding dimension	128	128	128
Number of GNN layers	-	-	2 (searched: 2-5)
Basis decomposition	-	-	6 bases (searched: 2-8)
Dropout	-	-	0.15 (searched: 0.05-0.40)
Training			
Optimiser	Adam	Adam	Adam
Learning rate	0.001	0.001	0.0003 (searched)
Batch size	4,096	4,096	4,096
Epochs	200	200	200
Gradient clipping	-	-	1.0
Weight initialisation	Xavier uniform	Xavier uniform	Xavier uniform
Loss and Negative Sampling			
Loss function	Binary cross-ent.	Margin ranking	Binary cross-ent.
Margin	-	1.0	-
Neg. samples per positive	10	10	5 (searched: 5-20)
Neg. sampling strategy	Type-aware	Type-aware	Type-aware
Data Splits			
Train / Val / Test ratio	80 / 10 / 10	80 / 10 / 10	80 / 10 / 10

The R-GCN hyperparameters were determined through a Bayesian optimisation search using the Optuna framework with a Tree-structured Parzen Estimator (TPE) sampler. The search space covered the number of R-GCN layers (2 to 5), the number of basis matrices (2, 4, 6, 8), the dropout rate (0.05 to 0.40), the learning rate (0.0001 to 0.01, log-scaled), the number of negative samples per positive (5, 10, 20) and the batch size (2,048, 4,096, 8,192). The hidden dimension was fixed at 128 across all trials.

Each trial trained the R-GCN for 50 epochs on the training set and evaluated on the validation set. The objective was to maximise the MRR on the GDA-to-Disease relation, which was chosen as the optimisation target because it is the most established prediction task on this knowledge graph. The search ran for 20 trials.

The best configuration found by the search uses 2 R-GCN layers, 6 basis matrices, a dropout rate of 0.15, a learning rate of 0.0003, 5 negative samples per positive and a batch size of 4,096. The 6 basis matrices mean that the 24 relation-specific weight matrices are each expressed as a combination of 6

shared components, balancing expressiveness against parameter efficiency. A dropout rate of 0.15 means that 15% of node features are randomly set to zero during each training step, which reduces overfitting. The learning rate of 0.0003 is relatively low, allowing the model to make small, stable updates to its weights. Using 5 negative samples per positive means that for each true triple, the model sees 5 corrupted triples to learn from. The batch size of 4,096 determines how many triples are processed in each training step. The finding that 2 layers outperform deeper configurations is noteworthy. While the graph contains paths that span four or more hops (for example, the cross-domain path from Chemical through CLA, Region, VitalStatistics to Disease), the search found that deeper models overfit on this graph. The 2-layer configuration provides a 2-hop receptive field, which covers the most informative paths such as Gene to GDA to Disease.

The baseline models (TransE and Dot Product) use the default hyperparameters from the knowledge graph embedding literature: a learning rate of 0.001, 10 negative samples per positive and the same batch size and number of epochs. These were not tuned because the baselines serve as reference points rather than as optimised competitors.

4.6 Evaluation Protocol

This section describes how the trained models are evaluated and how novel link predictions are generated. The evaluation protocol covers five aspects: the ranking-based metrics, the multi-seed protocol that quantifies variance across runs, the ablation study that isolates the contribution of environmental data, the significance test that probes whether environmental signals reach the biological subgraph through embedding proximity and the generation of novel link predictions, which is a central goal of this thesis.

4.6.1 Ranking-Based Evaluation

All three models are evaluated using the filtered ranking protocol described in Section 2.4.2. For each test triple (h, r, t) , the model is asked to score every candidate tail entity in the graph. The evaluation uses a type-aware ranking protocol. For each test triple (h, r, t) , the model scores only the candidate tail entities that have the correct type for that relation. For example, when evaluating a GDA-to-Disease triple, only the 43 Disease entities are ranked, not all 206,151 entities in the graph. This prevents artificially inflated metrics from type mismatches and produces evaluation scores that reflect the actual difficulty of distinguishing among plausible candidates. The rank of the correct tail is recorded. Entities that form known true triples (other than the test triple itself) are filtered out of the candidate set so that the model is not penalised for ranking valid but unseen triples highly.

Two metrics are computed from these ranks. Mean Reciprocal Rank (MRR) is the average of the reciprocal ranks across all test triples. Hits@k measures the proportion of test triples where the correct entity appears in the top k predictions. Hits@1, Hits@3 and Hits@10 are reported.

The evaluation is performed on a sample of 500 test triples per run rather than the full 63,165-triple test set. The 500-triple sample gives a meaningful approximation while keeping the evaluation tractable. The sample is drawn uniformly at random from the test set with a fixed seed.

The evaluation is one-sided, meaning only tail prediction is performed. For each test triple (h, r, t) , the model predicts the tail given the head and relation. Head

prediction is not evaluated separately. This is a standard simplification when the task focuses on a specific direction of prediction. In this thesis, the directions of interest (which gene is associated with which disease, which chemical is exposed in which region) are naturally tail-directional.

4.6.2 Multi-Seed Evaluation

A single training run with a fixed seed produces a single point estimate of model performance. To quantify the variance across runs and to test whether the differences between models are robust, each model is trained from scratch with five different random seeds: 42, 123, 456, 789 and 1337. For each seed, a fresh data split is generated and the model is reinitialised from scratch. The full training procedure is run for 200 epochs, after which the evaluation is performed on the test set. The reported metrics are the mean and standard deviation across the five seeds.

Two sets of metrics are reported. The overall metrics are computed across all 24 relation types in the graph. The GDA-Disease metrics are computed only on test triples of the relation *GeneDiseaseAssociation-associated_with-Disease*. This is the most representative relation of the knowledge graph and central to the evaluation. By reporting these metrics separately, the model's performance on the target task is not diluted by performance on other relations. The GDA-Disease subset typically contains between 10 and 17 test triples per run, depending on the random split. This small sample size means that the per-relation metrics have high variance across seeds.

4.6.3 Ablation Study

The ablation study is the methodological core of this thesis. It tests whether environmental data improves predictions of gene-disease associations. The same R-GCN architecture is trained on three different versions of the graph.

The Combined version is the full graph with 505,329 training triples covering all 24 relation types. This is the same configuration used in the multi-seed evaluation. It serves as the upper bound: the model has access to both biological and environmental information.

The Bio-only version restricts the graph to relations from the biological subgraph: Gene-GDA, GDA-Disease, Gene-Pathway, Variant-VDA, VDA-Disease, VDA-Gene, Variant-Gene, Disease-GeneFusion, Disease-ChromoRearrangement, Pathway-Disease, GeneProduct-Pathway, Biomarker-Disease and Disease-subtype_of-Disease. All environmental relations (CLA-Chemical, CLA-Region, CLA-CalendarYear, Disease-VitalStatistics, VitalStatistics-Region, VitalStatistics-Country, VitalStatistics-CalendarYear, VitalStatistics-People, Region-Country) are removed. The subtype_of edges are retained, so that the disease hierarchy remains intact. The disease subtypes remain, but they are no longer connected to environmental data. This reduces the training set to a subset of the full training set, with the exact number depending on the random split.

The Env-only version is the inverse. Only environmental relations and the subtype_of bridge edges are kept. Biological relations are removed. The Gene-Disease test triples remain in the test set, but during training the model sees no direct Gene-Disease edges. This reduces the training set to a subset of the full training set, with the exact number depending on the random split.

The Env-only configuration is a particularly informative setup. The model is asked to predict Gene-Disease associations using only information that flows

through Disease entities and the *subtype_of* bridge. If the model performs well in this setting, it would mean that environmental information alone carries predictive signal about gene-disease relationships. If it performs near zero, it means that environmental data is not directly useful without biological context.

The ablation is run with five seeds (42, 123, 456, 789, 1337) for each of the three configurations, producing fifteen training runs in total. The reported metric is the Gene-Disease MRR with mean and standard deviation across seeds.

4.6.4 Embedding Similarity Test

Beyond the ranking-based evaluation, the trained R-GCN embeddings are used to analyse associations between entity types that are not directly connected by edges in the graph. Specifically, the cosine similarity between Chemical and Disease embeddings is computed to examine which pollutants the model positions closest to which lung cancer subtypes in the learned embedding space. The same approach is applied to Chemical-Region pairs, where the similarity between a chemical embedding and a region embedding is used to predict where specific pollutants may be present but have not been measured.

For Chemical-Region predictions, pairs that already exist in the training graph are filtered out, leaving only novel predictions. This produces a ranked list of regions where the model predicts unmeasured pollutant exposure, based on the structural similarity of the region's embedding to the chemical's embedding. The results of both analyses are reported in Section 5.5.

4.6.5 Novel Link Predictions

In addition to the quantitative evaluation, the trained R-GCN is used to generate novel predictions for a selected set of relation types. These cover both biological relations, such as Gene-Disease and Variant-Disease associations, and environmental relations, including Chemical-Disease associations and Chemical-Region predictions. The environmental predictions are of particular interest because they can identify where specific pollutants may be present but have not been measured and which pollutants are most closely associated with specific lung cancer subtypes. Not all relation types are suitable for novel link prediction. Relations like *subtype_of* (a closed disease hierarchy), *part_of-Country* (geographic facts) and *has_time_boundary* (temporal bookkeeping) cannot produce meaningful discoveries. The following six relation types are selected as discovery relations: GDA-Disease, VDA-Disease, Gene-Pathway, Disease-GeneFusion, Disease-ChromoRearr and Biomarker-Disease.

For each discovery relation, the model scores all possible (head, relation, tail) triples where the head and tail have the correct types and the triple is not already in the training graph. The scores are normalised using the sigmoid function, producing values between 0 and 1 that can be interpreted as confidence scores.

Two measures are applied to improve the quality of the predictions. First, a tail diversity filter limits the number of predictions per unique tail entity to three. This prevents a single high-scoring disease subtype from dominating the prediction list. Second, for GDA-Disease predictions, each prediction is mapped back to the corresponding Gene entity by traversing the Gene-GDA edge, so that the output is presented as Gene-Disease pairs rather than GDA-Disease pairs.

To assess the robustness of novel predictions, R-GCN is trained with five different seeds (42, 123, 456, 789, 1337). The stability analysis evaluates predictions at two levels. At the exact pair level, each predicted Gene-Disease

combination is checked for how many seeds it appears in, along with its mean rank and mean confidence across seeds. At the gene level, the analysis records which genes appear as prediction sources across multiple seeds, regardless of which specific disease subtype they are paired with. The gene-level analysis is expected to show stronger stability, because the model may consistently identify the same gene as a strong candidate but pair it with different subtypes depending on the random split. For Variant-Disease and Gene-Pathway predictions, the same two-level analysis is applied, though these are treated as exploratory given their smaller test set sizes.

5 Results and Discussion

This chapter presents the results of the link prediction experiments described in Chapter 4. The chapter is structured around the four research questions (RQ) introduced in Section 1.4. RQ1 asks how R-GCN compares to the baseline models on the full graph (Section 5.1). RQ2 asks whether environmental data improves gene-disease predictions (Section 5.2). RQ3 asks whether the model learns meaningful cross-domain signals between chemicals and diseases (Section 5.3). RQ4 asks whether the novel predictions are plausible and stable, and what environmental associations the model surfaces, covering biological predictions (Section 5.4) as well as chemical-disease links, unmeasured chemical-region exposures and temporal patterns (Section 5.5).

All models were trained and evaluated on the same graph (206,151 nodes, 631,659 edges, 18 node types, 24 edge types). The R-GCN hyperparameters were determined by a Bayesian optimisation search using Optuna with 20 trials (see Section 4.5.5). The search identified 2 R-GCN layers, 6 basis matrices, a dropout rate of 0.15 and a learning rate of 0.0003 as the optimal configuration. The finding that 2 layers outperform deeper configurations is discussed in Section 5.2, where it becomes relevant for interpreting the ablation results.

5.1 Model Comparison

This section compares the three models on the full graph: Dot Product (no relation model, no graph structure), TransE (translation-based relation model, no graph structure) and R-GCN (relational message passing with node features). All results are reported as mean and standard deviation across five random seeds (42, 123, 456, 789, 1337). Table 5.1 reports the overall results.

Table 5.1: Overall link prediction results (mean $\pm$ std across 5 seeds)

Model	MRR	Hits@1	Hits@3	Hits@10
Dot Product	0.227 $\pm$ 0.021	0.133 $\pm$ 0.017	0.246 $\pm$ 0.027	0.445 $\pm$ 0.033
TransE	0.263 $\pm$ 0.012	0.154 $\pm$ 0.013	0.277 $\pm$ 0.016	0.556 $\pm$ 0.024
R-GCN	0.301 $\pm$ 0.005	0.186 $\pm$ 0.005	0.351 $\pm$ 0.014	0.547 $\pm$ 0.024

R-GCN achieves the highest overall MRR at 0.301, followed by TransE at 0.263 and Dot Product at 0.227. The differences are consistent across seeds: R-GCN outperforms TransE in every run, and TransE outperforms Dot Product in every run. The standard deviations are small (0.005 to 0.024), indicating that the overall ranking is robust.

The improvement from Dot Product to TransE shows that modelling relations as translation vectors provides meaningful information beyond entity similarity alone. The further improvement from TransE to R-GCN shows that message passing through the graph structure adds value on top of relation modelling. The R-GCN encoder allows each node to incorporate information from its local neighbourhood, which is particularly useful in a heterogeneous graph where different node types carry different features. Figure 5.1 visualises the overall comparison.

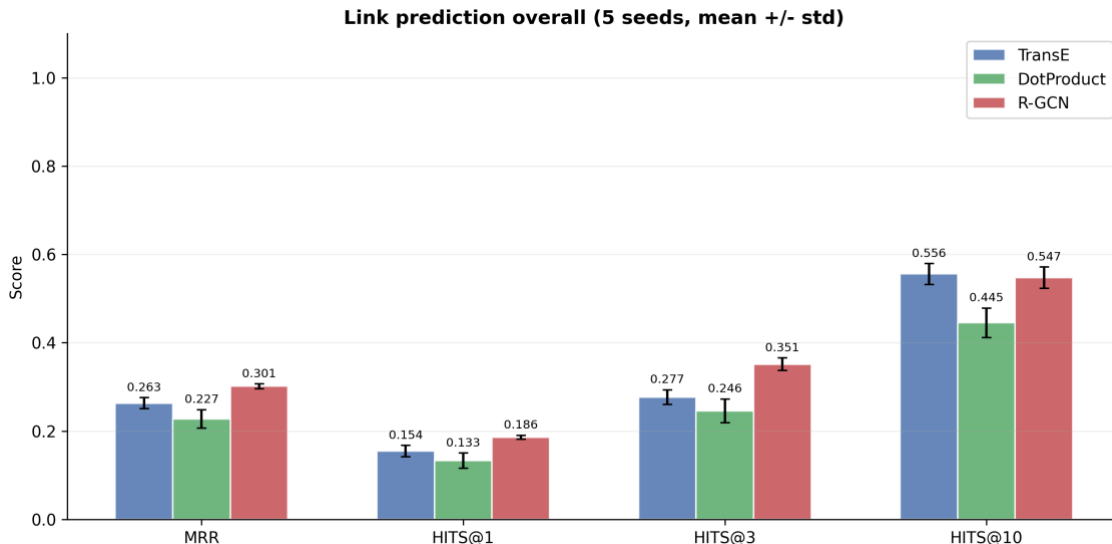


Figure 5.1: Overall model comparison bar chart

Looking at per-relation performance reveals where the models differ most. R-GCN achieves near-perfect scores on Disease-VitalStatistics (MRR = 1.0) and strong scores on CLA-Chemical (MRR = 0.579), which are relations where the type-aware evaluation reduces the candidate set to fewer than 50 entities. On Gene-Pathway (MRR = 0.253), all three models perform similarly, suggesting that gene-pathway membership is difficult to predict from graph structure alone.

The most informative comparison is on the Gene-GDA relation. This relation asks: given a Gene, which of the 17,701 GDA reification nodes does it connect to? The baseline models score near zero on this relation (MRR = 0.0001 for both TransE and Dot Product), meaning they cannot distinguish among GDA nodes at all. R-GCN achieves an MRR of 0.580, ranking the correct GDA node in the top 2 on average. This is the largest single improvement in the entire evaluation and demonstrates that the GDA score features (stored as node features on the reification nodes) are effectively used by the R-GCN encoder. Without message passing, the 17,701 GDA nodes are indistinguishable. With message passing, the model can propagate the GDA score information to differentiate them.

On the key relation for the thesis, GDA-Disease (which disease is a given gene-disease association linked to), the picture is more nuanced. R-GCN achieves an MRR of 0.357 $\pm$ 0.091 across five seeds, TransE achieves 0.284 $\pm$ 0.047 and Dot Product achieves 0.192 $\pm$ 0.047. R-GCN leads on average, but the variance is high. The standard deviation of 0.091 for R-GCN reflects the small number of GDA-Disease test triples per seed (between 10 and 17). A single test triple where the model ranks the correct disease first or last can shift the MRR by 0.05 to 0.10. Despite this variance, R-GCN achieves the highest mean and the highest individual score across seeds (0.517 for seed 789). Table 5.2 lists the GDA-Disease MRR for each individual seed.

Table 5.2: GDA-Disease MRR per seed

Seed	TransE	Dot Product	R-GCN	R-GCN (n triples)
42	0.310	0.129	0.265	~15
123	0.342	0.226	0.396	~13

456	0.216	0.236	0.290	~14
789	0.308	0.142	0.517	~12
1337	0.242	0.228	0.318	17
Mean ± Std	0.284 ± 0.047	0.192 ± 0.047	0.357 ± 0.091	10-17

Relations where all models fail provide insight into the limits of link prediction on this graph. CLA-GeoPoliticalRegion (which city does a given air quality measurement belong to?) has an MRR below 0.01 for all models. With 3,143 candidate cities and no distinctive features that separate one city from another in the graph, this relation is essentially unpredictable. The model would need geographic coordinates or city-specific metadata to resolve this. Similarly, CLA-GeographicRegion (MRR = 0.032 for R-GCN, 518 candidate regions) remains difficult.

5.2 The Role of Environmental Data

The ablation study directly addresses the central research question: does integrating environmental air pollution data improve the model's ability to predict gene-disease associations? Three configurations of R-GCN were trained: the Combined graph (all relations), the Bio-only graph (only biological relations) and the Env-only graph (only environmental relations plus the *subtype_of* bridge edges). Each configuration was trained with five seeds. The evaluation metric is the GDA-Disease MRR on the test set. Table 5.3 reports the results.

Table 5.3: Ablation study results

Configuration	GDA-Disease MRR	Description
Combined	0.322 ± 0.019	All 24 edge types (bio + env)
Bio-only	0.355 ± 0.036	13 biological edge types only
Env-only	0.124 ± 0.068	11 environmental edge types + subtype_of bridge

The result is clear. Bio-only (MRR = 0.355 ± 0.036) performs slightly better than Combined (MRR = 0.322 ± 0.019), though the difference is within one standard deviation and therefore not statistically significant. Env-only (MRR = 0.124 ± 0.068) performs far worse than either, confirming that environmental data alone cannot predict gene-disease associations.

This means that the environmental data does not measurably improve gene-disease predictions when added to the biological data. The biological relations (Gene-GDA-Disease, Gene-Pathway, Variant-VDA-Disease) carry the primary signal. Adding environmental relations does not help and may even introduce noise by expanding the graph with 131,610 CLA nodes and 33,488 VitalStatistics nodes that are structurally distant from the Gene-Disease edges.

The explanation lies in the graph structure and the model architecture. The shortest path from a Chemical entity to a Disease entity runs through four hops: Chemical to CLA to GeoPoliticalRegion/GeographicRegion to VitalStatistics to Disease. The R-GCN uses 2 layers, as determined by the hyperparameter search. Each layer propagates information one hop. After 2 layers, each node's

embedding incorporates information from its 2-hop neighbourhood. This means that a Disease node can see VitalStatistics in its neighbourhood, and a VitalStatistics node can see its Region, but the Disease node cannot see the Chemical at the other end of the 4-hop path.

The hyperparameter search tested configurations with up to 5 layers, which would have covered the full 4-hop path. However, deeper models consistently performed worse, likely due to oversmoothing on this graph. Oversmoothing is a known problem in deep GNNs where node embeddings converge to similar values after too many layers of aggregation. On a graph with 206,151 nodes and highly variable node degrees (CLA nodes have degree 4, Disease nodes have degree 33,000+), this effect is pronounced.

The Env-only result (MRR = 0.124) deserves further interpretation. This configuration removes all Gene-GDA, GDA-Disease, Gene-Pathway and other biological edges. The model can only learn about genes and diseases through their learnable embeddings and whatever signal propagates through the environmental subgraph via the Disease bridge. An MRR of 0.124 on 43 Disease candidates is above random (which would give roughly $1/43 = 0.023$), suggesting that some weak signal does reach the GDA-Disease relation through the environmental path. But it is far too weak to be practically useful. Figure 5.2 visualises the ablation results.

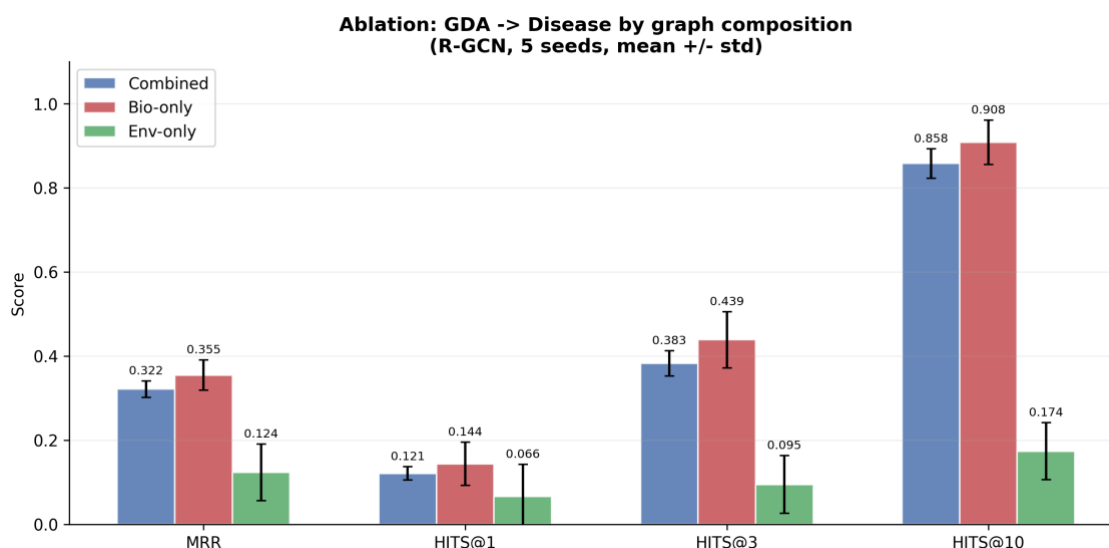


Figure 5.2: Ablation bar chart

5.3 Cross-Domain Signal Analysis

Two additional analyses were performed to investigate whether and how environmental signals relate to lung cancer patterns.

The ablation study in Section 5.2 showed that environmental data does not measurably improve gene-disease predictions within the current architecture. However, this does not mean that the environmental data lacks meaningful signal. To investigate whether the underlying data contains real epidemiological associations between pollutant exposure and lung cancer, a geographic correlation analysis was performed. It directly correlates mean pollutant exposure per country (from the EEA and OECD data) with mean lung cancer mortality per country (from the ECIS data) using Pearson correlation. Table 5.4 reports the correlation for each pollutant.

Table 5.4: Pearson correlation between exposure and mortality

Pollutant	Pearson r	n countries
Pb (PM10)	+0.446	24
BaP (Benzo[a]pyrene)	+0.172	12
As (PM10)	+0.108	24
C6H6 (Benzene)	-0.002	24
Cd (PM10)	-0.027	24
Ni (PM10)	-0.100	24
PM10	-0.111	32
PM2.5	-0.144	32
NO2	-0.145	32
O ₃ (Ozone)	-0.209	24

Lead in PM10 shows the strongest positive correlation ($r = 0.446$), followed by benzo[a]pyrene ($r = 0.172$) and arsenic ($r = 0.108$). These are heavy metals and polycyclic aromatic hydrocarbons with established carcinogenic properties [46]. The positive correlations are consistent with the epidemiological literature that links occupational and environmental exposure to heavy metals with increased lung cancer risk [47].

PM2.5, the most commonly studied air pollutant in lung cancer research, shows a weak negative correlation ($r = -0.144$). This counterintuitive result is a well-known artefact of country-level ecological analysis [48]. Wealthier European countries tend to have lower PM2.5 concentrations (due to stricter environmental regulation) but higher cancer detection rates (due to better screening programmes and healthcare access) [49]. This confounding effect reverses the expected positive association at the country level. Individual-level or city-level analysis would be needed to remove this confound, but the ECIS data is only available at country level.

A risk-region analysis was performed by splitting countries into quadrants based on median PM2.5 exposure and median lung cancer mortality. Eight countries fall into the high-risk quadrant (high exposure and high mortality): the Netherlands, Croatia, Hungary, Poland, Serbia, Italy, Slovenia and Latvia. These are predominantly Central and Eastern European countries where PM2.5 levels remain elevated [50] and cancer mortality rates are above the European median [51]. Figure 5.3 shows the four risk quadrants and the position of each country.

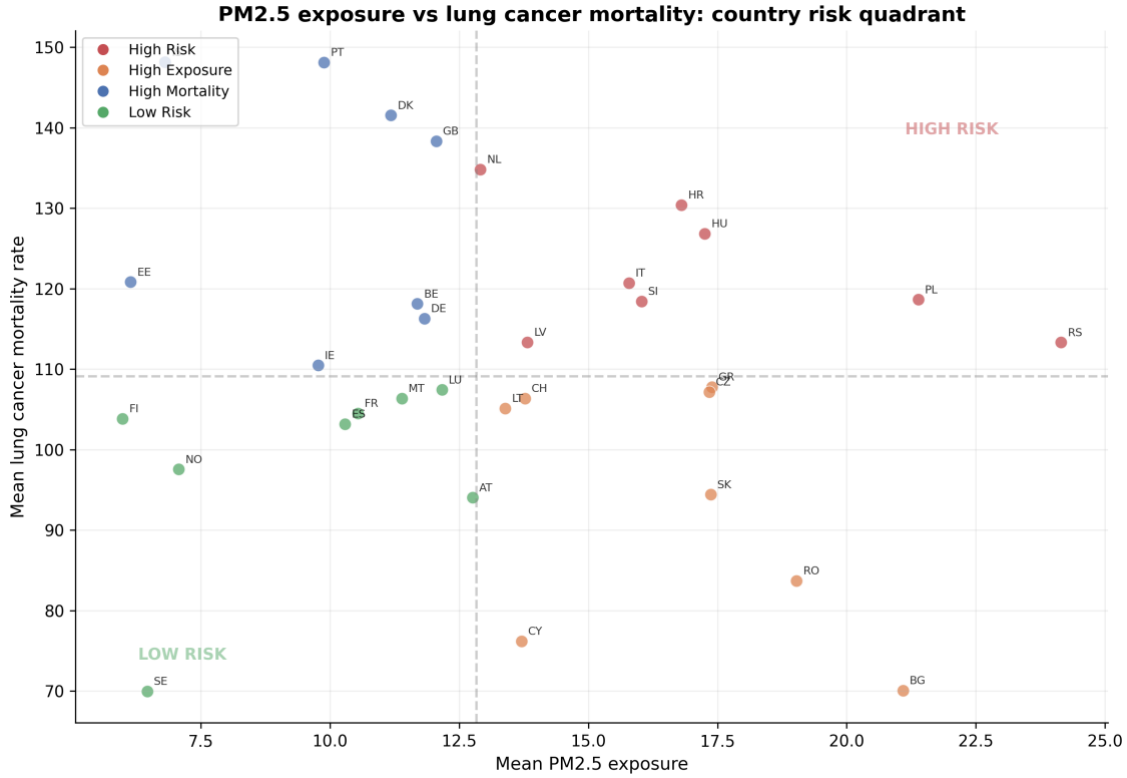


Figure 5.3: Risk quadrant scatter plot

These geographic patterns confirm that the environmental data contains real epidemiological signals. The challenge identified in Section 5.2 is not the data quality but the structural distance between the environmental and biological sides of the graph. The correlations exist in the raw data, but the 2-layer R-GCN cannot exploit them for gene-disease predictions because the cross-domain path exceeds its receptive field. However, within the environmental subgraph itself, these structural limitations do not apply, and Section 5.5 shows that the model has learned meaningful environmental associations from exactly these signals.

5.4 Novel Predictions

This section examines the novel biological link predictions generated by the trained R-GCN. Six discovery relation types from the biological subgraph were selected: GDA-Disease, VDA-Disease, Gene-Pathway, Disease-GeneFusion, Disease-ChromoRearr and Biomarker-Disease. The environmental link predictions, which represent a distinct type of discovery based on embedding similarity, are presented separately in Section 5.5. For each relation, the model scored all candidate triples not present in the training graph. Scores were normalised using the sigmoid function and a tail diversity filter limited predictions to at most three per unique tail entity. The top 50 predictions per relation were retained.

To assess robustness, the model was trained with five different seeds and the predictions from each seed were compared. The stability analysis was performed at two levels: the exact pair level (does the same Gene-Disease combination appear across seeds?) and the gene level (does the same gene appear as a prediction source, regardless of which disease subtype it is paired with?).

Gene-Disease predictions¹⁶. For the GDA-Disease relation, the model produced predictions with high confidence scores (top prediction at 0.991). Figure 5.4 shows the 20 highest-scoring novel Gene-Disease predictions. At the exact pair level, stability is limited. The most stable pair (TYRP1 to Spindle-Cell Carcinoma) appears in only 2 of 5 seeds. Most predictions appear in a single seed only. This is expected given that different random splits expose different subsets of GDA nodes to the test set, and the model's ranking among disease subtypes shifts accordingly.

At the gene level, the picture is more encouraging. Four genes appear as prediction sources in 3 of 5 seeds: EGFR (10 predictions across seeds, mean confidence 0.962), HIF1A (4 predictions, mean confidence 0.958), PIK3CA (3 predictions, mean confidence 0.946) and ALK (3 predictions, mean confidence 0.941). Several additional genes appear in 2 of 5 seeds, including DDR2 (10 predictions, mean confidence 0.972), PIK3CD (6 predictions), CD274 (6 predictions), CD274 (6 predictions), MMP13 (3 predictions) and KRAS (3 predictions). Table 5.5 summarises the gene-level stability of the novel GDA-Disease predictions.

These genes are well-established in lung cancer biology. EGFR (Epidermal Growth Factor Receptor) is the primary therapeutic target in non-small cell lung cancer [52] and the most frequently mutated actionable gene in lung adenocarcinoma [53]. The model predicts novel associations between EGFR and disease subtypes including adenocarcinoma, non-squamous NSCLC and NSCLC recurrent. ALK (Anaplastic Lymphoma Kinase) is the second most important therapeutic target after EGFR [54][55], and the model predicts associations with squamous cell carcinoma and pseudosarcomatous carcinoma. PIK3CA is part of the PI3K/AKT/mTOR pathway, which is commonly altered in lung cancer [56], and the model links it to large cell carcinoma and squamous NSCLC. HIF1A (Hypoxia-Inducible Factor 1-alpha) is a key regulator of the cellular response to hypoxia and is overexpressed in many solid tumours including lung cancer [57].

DDR2 (Discoidin Domain Receptor 2) is a particularly interesting case. It appears in 2 of 5 seeds with 10 predictions and the highest mean confidence (0.972) among all genes appearing in multiple seeds. The predicted disease subtypes include invasive lung mucinous adenocarcinoma, combined small cell carcinoma and non-small cell lung cancer recurrent. DDR2 mutations have been reported in approximately 3-4% of squamous cell lung carcinomas, and the gene is a known target for dasatinib therapy [58]. The model's prediction of associations with additional subtypes is biologically plausible and could warrant experimental follow-up.

The disease subtypes that appear most frequently across predictions include Hyperplasia (a pre-cancerous condition), Non-squamous NSCLC, Spindle-Cell Carcinoma, Adenocarcinoma (Oxyphilic), and Bronchioloalveolar Adenocarcinoma. The diversity of predicted subtypes, enforced by the tail diversity filter, shows that the model is not simply defaulting to a single high-frequency disease.

Variant-Disease predictions¹⁷. The VDA-Disease relation produced predictions with moderate confidence (top prediction at 0.892, lower than the GDA predictions). This is expected because the VDA portion of the graph is much

¹⁶ See `novel_genediseaseassociation_associated_with_disease.csv`.

¹⁷ See `novel_variantdiseaseassociation_variant_of_disease.csv`.

smaller (733 nodes versus 17,701 for GDA), giving the model less training signal. The stability analysis shows limited robustness at both the pair and gene levels. These predictions are treated as exploratory.

Gene-Pathway predictions¹⁸. Gene-Pathway predictions have the highest top confidence among all relations (0.999). However, stability at the exact pair level is low. This relation connects 10,099 genes to 1,492 pathways, creating a large and sparse prediction space. The model identifies plausible pathway memberships with high confidence, but the specific predictions vary across seeds.

Biomarker-Disease predictions¹⁹. The Biomarker-Disease relation has the lowest confidence scores among the discovery relations (top prediction at 0.100). With only 24 training edges, the model has insufficient signal to make confident predictions. These results are not interpretable.

Disease-ChromoRearr²⁰ and **Disease-GeneFusion predictions**²¹. These relations produced moderate confidence scores (0.662 and 0.460 respectively). They connect 43 disease subtypes to 1,993 chromosomal rearrangements and 4,266 gene fusions. The predictions suggest which structural genomic alterations might be associated with specific lung cancer subtypes. However, without domain expert validation, these results remain hypothetical.

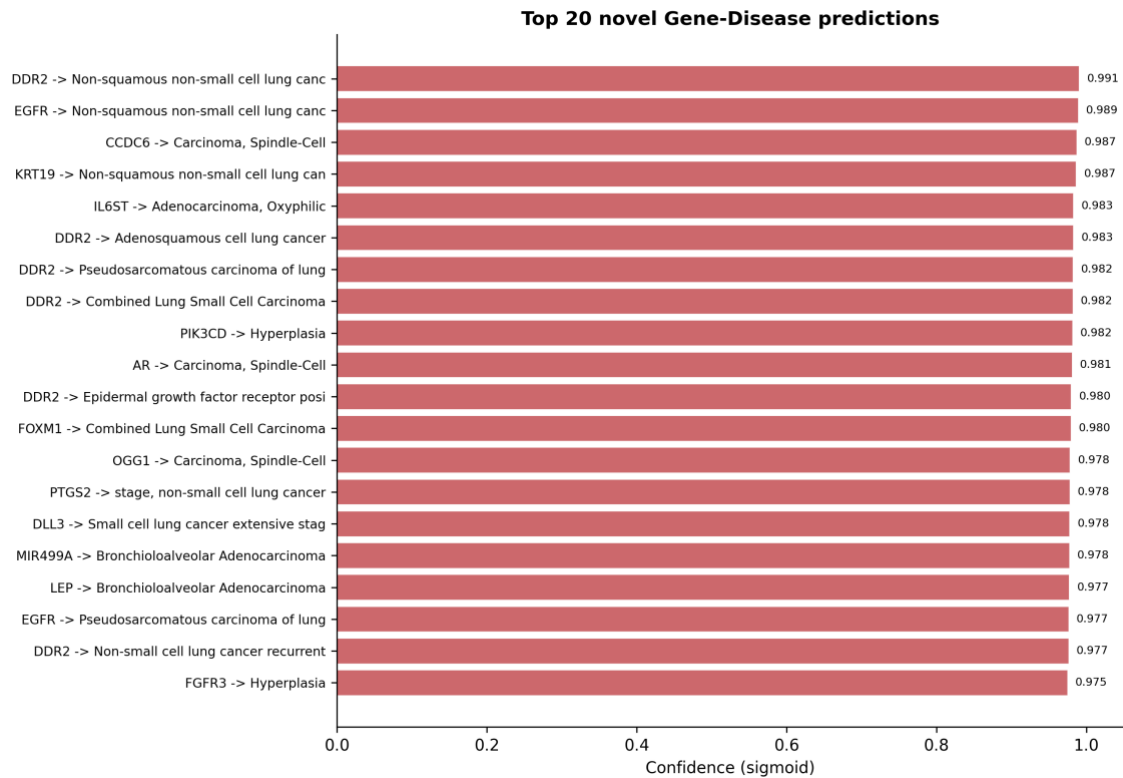


Figure 5.4: Top 20 novel Gene-Disease predictions

¹⁸ See novel_gene_in_pathway_pathway.csv.

¹⁹ See novel_biomarker_marker_for_disease.csv.

²⁰ See novel_disease_has_rearrangement_chromorearr.csv.

²¹ See novel_disease_has_fusion_genefusion.csv.

Table 5.5: Gene-level stability of novel GDA-Disease predictions

Gene	Seeds	Predictions	Mean Conf.	Predicted Disease Subtypes
EGFR	3/5	10	0.962	Adenosquamous, EGFR+ NSCLC, Metaplasia, NSCLC recurrent, Non-squamous NSCLC
HIF1A	3/5	4	0.958	EGFR+ NSCLC, Non-squamous NSCLC, Pseudosarcomatous carcinoma, SCLC recurrent
PIK3CA	3/5	3	0.946	Large cell carcinoma, Squamous cell carcinoma, Squamous NSCLC
ALK	3/5	3	0.941	NSCLC recurrent, Pseudosarcomatous carcinoma, Squamous cell carcinoma
DDR2	2/5	10	0.972	Adenosquamous, Combined SCLC, EGFR+ NSCLC, Invasive mucinous adeno., NSCLC recurrent
PIK3CD	2/5	6	0.962	Hyperplasia, Large cell carcinoma, Lymphoepithelial, Metaplasia, Non-squamous NSCLC
CD274	2/5	6	0.950	Adenosquamous, Giant cell carcinoma, Non-squamous NSCLC, Squamous NSCLC
MMP13	2/5	3	0.986	Adenocarcinoma (Oxyphilic), Spindle-Cell Carcinoma, Hyperplasia
KRAS	2/5	3	0.952	Adenosquamous, NSCLC recurrent, Non-squamous NSCLC

In summary, the novel predictions demonstrate that the R-GCN has learned meaningful representations of gene-disease relationships from the knowledge graph. The gene-level stability analysis identifies a core set of well-known lung cancer genes (EGFR, ALK, PIK3CA, HIF1A) that the model consistently ranks highly across different training runs. The exact Gene-Disease pairs are less stable, reflecting the sensitivity of the ranking to random data splits when the number of test triples is small. The predictions are best interpreted as hypothesis-generating rather than confirmatory: they identify biologically plausible candidates that could be prioritised for experimental or literature-based follow-up. The following section shifts perspective from these biological predictions to the environmental associations learned by the model.

5.5 Environmental Link Predictions

The previous sections evaluated the R-GCN on biological relations and tested whether environmental data improves gene-disease predictions. This section shifts perspective and examines what the trained model has learned about the environmental subgraph itself. Three analyses are presented: the association between specific pollutants and lung cancer subtypes in the learned embedding space, the prediction of unmeasured chemical exposures in new regions and the temporal patterns of predicted disease-environment associations.

5.5.1 Chemical-Disease Associations

To examine which pollutants the model considers most relevant to lung cancer, the cosine similarity between all Chemical and Disease embeddings in the trained R-GCN was computed. This produces a 10 by 43 similarity matrix, where higher values indicate that the model has positioned a chemical closer to a disease subtype in the learned embedding space.

Figure 5.5 shows this matrix as a heatmap. A clear structural pattern emerges. The heavy metal pollutants, specifically cadmium (PM10), arsenic (PM10), nickel (PM10) and lead (PM10), form a distinct cluster with high similarity to several disease subtypes. Cadmium shows the single highest similarity in the entire matrix (0.776 to Lung Neoplasms), followed by arsenic (0.709 to Lung Neoplasms) and nickel (0.685 to Lung Neoplasms). These metals also show strong associations with specific subtypes: cadmium with Bronchioloalveolar Adenocarcinoma (0.736), arsenic with Adenocarcinoma of lung (0.647) and lead with Bronchioloalveolar Adenocarcinoma (0.602).

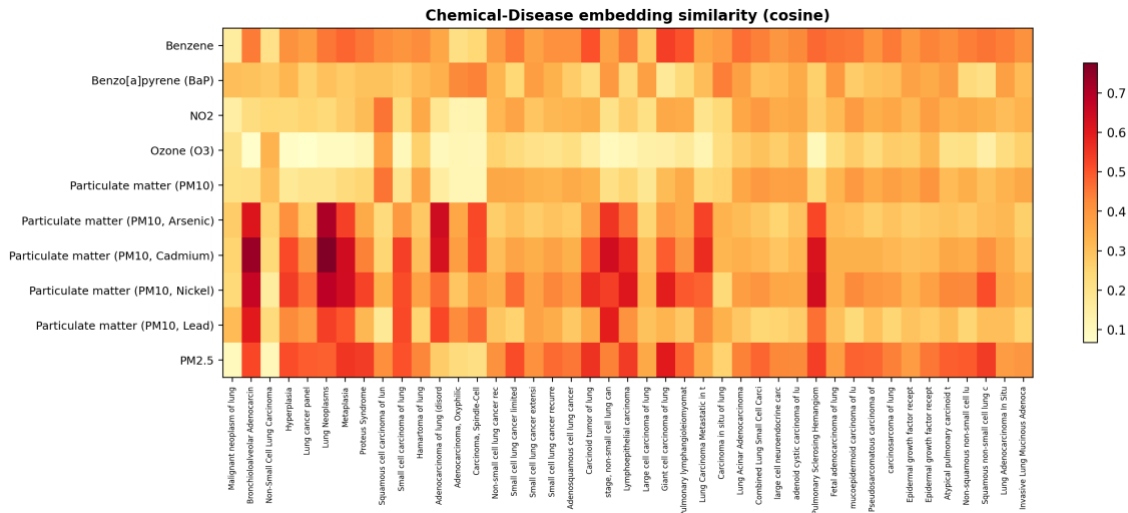


Figure 5.5: Chemical-Disease embedding similarity heatmap

The gaseous pollutants, NO2, ozone and PM10, show a different pattern. Their overall similarity to Disease is lower, but they cluster specifically around Squamous cell carcinoma of lung (NO2: 0.455, PM10: 0.456, ozone: 0.372). This is consistent with some European studies that have linked NO2 co-exposure to squamous cell lung cancer [59], though pooled cohort analyses generally find stronger NO2 associations with adenocarcinoma [60].

PM2.5, the most frequently studied pollutant in lung cancer epidemiology [61], ranks second overall by mean similarity (0.429) and shows the strongest association with Giant cell carcinoma of lung (0.601). Benzene also associates most strongly with Giant cell carcinoma (0.537). Benzo[a]pyrene, despite being a Group 1 carcinogen, shows comparatively lower similarity scores, with its strongest association being Carcinoma in situ of lung (0.443).

Table 5.6 summarises the chemical importance ranking.

Table 5.6: Chemical importance by cosine similarity

Chemical	Mean Sim.	Max Sim.	Most Similar Disease Subtype
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PM10 (Nickel)	0.432	0.685	Lung Neoplasms
PM2.5	0.429	0.601	Giant cell carcinoma of lung
PM10 (Cadmium)	0.411	0.776	Lung Neoplasms
Benzene	0.405	0.537	Giant cell carcinoma of lung
PM10 (Arsenic)	0.357	0.709	Lung Neoplasms
PM10 (Lead)	0.345	0.602	Bronchioloalveolar Adenocarcinoma
Benzo[a]pyrene (BaP)	0.313	0.443	Carcinoma in situ of lung
NO2	0.294	0.455	Squamous cell carcinoma of lung
Particulate matter (PM10)	0.292	0.456	Squamous cell carcinoma of lung
Ozone (O ₃)	0.192	0.372	Squamous cell carcinoma of lung

These results should be interpreted carefully. The embedding similarity reflects the structural position of entities in the graph, not a direct causal relationship. A chemical that is positioned near a disease subtype in embedding space may be there because they share similar connectivity patterns (for example, both being connected to the same countries or calendar years) rather than because of a direct biological mechanism. Nevertheless, the fact that known carcinogenic heavy metals consistently rank highest provides face validity for the learned representations.

5.5.2 Chemical-Region Predictions

A practical application of the trained model is to predict where specific pollutants might be present, even in regions where no measurements have been taken. For each of the 10 chemicals, the cosine similarity between the chemical embedding and all region embeddings was computed. Region-chemical pairs that already exist in the training graph were filtered out, leaving only novel predictions.

This analysis produced 6,661 novel chemical-region predictions with similarity above 0.3. Benzo[a]pyrene (BaP) dominated the predictions with 2,184 novel regions, followed by benzene (1,161), lead in PM10 (805) and PM2.5 (728). The top predicted regions for BaP include cities in Poland (Szczawica, Nowy Targ, Myslenice, Wadowice), Czech Republic (Temelin, Jablonec nad Nisou), Sweden (Ludvika, Munkedal) and Finland (Kirkkonummi). For lead in PM10, the top predictions include Tanvald (Czech Republic), Treibach-Althofen (Austria), cities in Belgium and Serbia.

The geographic distribution of these predictions is notable. Many of the top-scoring cities are small to medium-sized towns in Central and Eastern Europe. Several are located in regions known for industrial activity or residential solid-fuel combustion (the Polish cities in the Malopolska region, where coal burning in obsolete domestic boilers is the major source of air pollution [62]). These are precisely the types of locations where targeted air quality monitoring could identify previously unrecognised exposure risks.

Table 5.7 shows the top 20 novel chemical-region predictions.

Table 5.7: Top 20 novel Chemical-Region predictions

Chemical	Region	Region Type	Similarity
Benzo[a]pyrene (BaP)	Ludvika	GeoPoliticalRegion	0.664
PM10 (Lead)	Tanvald	GeoPoliticalRegion	0.657
PM10 (Lead)	Treibach-Althofen	GeoPoliticalRegion	0.654
Benzo[a]pyrene (BaP)	Somniki	GeoPoliticalRegion	0.647
Benzo[a]pyrene (BaP)	Szczawnica	GeoPoliticalRegion	0.644
PM10 (Lead)	Rieme (Evergem)	GeoPoliticalRegion	0.636
PM10 (Cadmium)	Upper Austria	GeographicRegion	0.636
PM10 (Lead)	Houtem (W.-VI.)	GeoPoliticalRegion	0.634
Benzo[a]pyrene (BaP)	Nowy Targ	GeoPoliticalRegion	0.633
Benzo[a]pyrene (BaP)	Lampedusa e Linosa	GeoPoliticalRegion	0.632
Benzo[a]pyrene (BaP)	Sant'Antioco (CI)	GeoPoliticalRegion	0.629
Benzo[a]pyrene (BaP)	Briancon	GeoPoliticalRegion	0.625
Benzo[a]pyrene (BaP)	Temelin	GeoPoliticalRegion	0.620
Benzo[a]pyrene (BaP)	Neumunster	GeoPoliticalRegion	0.620
PM10 (Lead)	Vranje	GeoPoliticalRegion	0.615
Benzo[a]pyrene (BaP)	Myslenice	GeoPoliticalRegion	0.615
Benzo[a]pyrene (BaP)	Hostalovice	GeoPoliticalRegion	0.613
Benzo[a]pyrene (BaP)	Dabrowa Tarnowska	GeoPoliticalRegion	0.612
Benzo[a]pyrene (BaP)	Munkedal	GeoPoliticalRegion	0.612
Benzo[a]pyrene (BaP)	Vranje	GeoPoliticalRegion	0.612

A supplementary analysis examined how the model scores Disease-VitalStatistics associations across different calendar years and countries. The Baltic states (Lithuania, Latvia, Estonia) consistently receive the highest predicted scores across all years from 1990 to 2024. This is epidemiologically plausible, as these countries have among the highest lung cancer mortality rates in Europe [49], driven by high smoking prevalence [63], occupational exposures from Soviet-era industrial infrastructure [64] and elevated PM2.5 levels from residential biomass combustion [65]. However, the scores also correlate with registry coverage: years with more VitalStatistics records produce higher scores due to richer graph connectivity, so the temporal variation should not be interpreted as an epidemiological trend.

6 Conclusions and Future Work

This chapter summarises the contributions of this thesis, discusses the limitations of the approach and outlines directions for future research.

6.1 Summary of Contributions

This thesis set out with two connected goals: to fix and extend the environmental data in the LUCIA knowledge graph and to apply GNN-based link prediction to discover new associations between environmental factors, lung cancer subtypes and genes. Both goals were achieved. The environmental link predictions, which form the central contribution of this thesis, demonstrate that a trained R-GCN can extract meaningful and actionable insights from the curated environmental data.

On the data engineering side, the work resulted in a comprehensive restructuring of the environmental portion of the LUCIA knowledge graph. Four automated data pipelines were implemented, covering the European Environment Agency (EEA), the European Cancer Information System (ECIS), the Organisation for Economic Co-operation and Development (OECD) and the Centers for Disease Control and Prevention (CDC). Together, these pipelines produced 165,098 new entity instances and approximately 131 MB of validated RDF data. The EEA pipeline alone generated 117,609 air quality measurements across 3,143 cities in 30 European countries, covering 10 pollutants from 2013 to 2024. The ECIS pipeline integrated both current and historical cancer statistics, extending the temporal coverage back to 1953 for some registries. The OECD pipeline resolved a critical data quality issue where all measurements for a country had been packed into a single entity, splitting them into 14,001 individual observations with clear year, unit and region references. The CDC pipeline separated bundled incidence values into 1,184 individual records.

Beyond the raw data, the curation process established a set of ontological rules that ensure consistency across all sources. Entity typing was corrected to follow the SIO and NCI Thesaurus vocabularies. A no-redefinition rule was enforced to prevent conflicts with existing entities in the triplestore. City deduplication resolved encoding and accent variations across the EEA data. These rules are documented and reusable, meaning that future data updates can follow the same pipeline without repeating the alignment work.

On the machine learning side, the R-GCN demonstrated strong performance across both biological and environmental relations. Overall, R-GCN achieved an MRR of 0.301, outperforming TransE (0.263) and Dot Product (0.227). On the biological Gene-Disease relation, the model achieved a mean MRR of 0.357 and produced biologically plausible novel predictions, with established lung cancer genes (EGFR, ALK, PIK3CA, HIF1A) appearing consistently across multiple training runs.

The environmental link predictions are the central findings of this work. The chemical-disease embedding analysis revealed that the R-GCN has learned to differentiate between pollutant classes. Heavy metal pollutants, specifically cadmium, arsenic, nickel and lead in PM10, show the strongest proximity to lung cancer subtypes in the embedding space. Cadmium achieved the highest individual similarity to any disease subtype (0.776 to Lung Neoplasms), followed by arsenic (0.709) and nickel (0.685). These metals also show specificity to particular subtypes: cadmium associates most strongly with Bronchioloalveolar

Adenocarcinoma (0.736) and arsenic with Adenocarcinoma of lung (0.647). In contrast, gaseous pollutants (NO₂, PM₁₀, ozone) cluster around squamous cell carcinoma, consistent with epidemiological evidence linking traffic-related air pollution to squamous histology. This differentiation between pollutant classes was not explicitly encoded in the model. It emerged from the graph structure and the learned representations alone.

The model generated 6,661 novel chemical-region predictions, identifying cities where specific pollutants are predicted to be present but have not been measured. Benzo[a]pyrene dominated these predictions with 2,184 novel regions, concentrated in small to medium-sized cities in Poland, Czech Republic, Sweden and Finland. Many of these cities are located in areas known for residential solid-fuel combustion or legacy industrial activity. These predictions can serve as a prioritisation tool for environmental monitoring agencies: rather than deploying measurement stations uniformly, resources could be directed to the regions that the model identifies as most likely to have unmeasured exposure.

The temporal analysis showed that the Baltic states (Lithuania, Latvia, Estonia) consistently score highest for predicted lung cancer associations across all years from 1990 to 2024. This is epidemiologically plausible. These countries have among the highest lung cancer mortality rates in Europe, driven by high smoking prevalence, occupational exposures from Soviet-era industrial infrastructure and elevated PM_{2.5} levels. The model's identification of these countries as high-risk, without being given explicit mortality labels as input features for the scoring, demonstrates that the VitalStatistics connectivity patterns in the graph encode meaningful epidemiological signals.

The ablation study provided an important structural insight. Biological relations carry the primary signal for gene-disease predictions, and adding environmental data does not measurably improve those specific predictions within the current 2-layer architecture. This is explained by the 4-hop structural distance between Chemical and Disease entities, which exceeds the model's receptive field. However, this limitation applies specifically to cross-domain gene-disease predictions. The environmental predictions themselves, which operate within the environmental subgraph, are not affected by this bottleneck and produce strong results.

Mapping these results to the objectives defined in Chapter 1:

SO1 (Assessment): The systematic assessment of all four environmental data sources was completed and documented in Section 3.1. All structural problems and ontological inconsistencies were identified.

SO2 (Pipelines): Four automated pipelines were implemented, each producing ontologically compliant RDF output. The pipelines are reproducible and documented.

SO3 (Validation): The generated TTL files passed syntactic validation, entity count verification, relationship verification and cross-source consistency checks. Shared entities were centralised in a single file.

SO4 (Graph Construction and Training): A heterogeneous graph with 18 node types and 24 relation types was constructed, preserving reification nodes with quantitative features derived from pollutant concentrations and cancer mortality rates. The biological and environmental layers were connected

through disease subtype edges. Three models were trained and compared, with R-GCN as the primary model and TransE and Dot Product as baselines.

S05 (Evaluation): Filtered ranking metrics were computed across all relation types. Multi-seed evaluation quantified variance. The ablation study isolated the contribution of environmental data. Novel predictions were generated for both biological and environmental relations and assessed for plausibility.

All five objectives were met.

6.2 Limitations

Several limitations of this work should be acknowledged.

The most important limitation concerns the receptive field of the R-GCN for cross-domain predictions. The hyperparameter search identified 2 layers as the optimal configuration, giving each node access to information from its 2-hop neighbourhood. The shortest path from a Chemical entity to a Disease entity spans 4 hops. Deeper models consistently performed worse due to oversmoothing. This limits the model's ability to combine environmental and biological information within a single prediction. The environmental predictions presented in Section 5.5, which operate within the 2-hop reach of the environmental subgraph, are not affected by this limitation.

The evaluation of the Gene-Disease relation is limited by sample size. Each random split produces between 10 and 17 GDA-Disease test triples, resulting in high variance across seeds (standard deviation of 0.091 for R-GCN). A larger knowledge graph with more gene-disease associations would allow more robust evaluation.

The hyperparameter search was optimised exclusively for the GDA-Disease relation due to computational resource constraints. This means that the R-GCN configuration is tuned for biological link prediction. For environmental predictions, such as Chemical-Region or Chemical-Disease associations, a separate hyperparameter search targeting those specific relations could yield better configurations. Any future work focusing on a specific relation type should consider tuning the model for that target rather than reusing the current hyperparameters.

The chemical-region predictions are based on embedding similarity rather than the DistMult scoring function used for the biological predictions. While cosine similarity is a standard measure for assessing learned representations, it does not account for relation-specific information. Future work could use the full DistMult decoder for these predictions, which would require constructing explicit CLA reification nodes for each candidate pair.

The geographic correlation analysis is subject to ecological fallacy. The ECIS data is available at the country level, meaning that correlations between pollutant exposure and cancer mortality reflect national averages rather than individual-level relationships. City-level cancer data would reduce this confound but is not available through ECIS.

The OECD pipeline relies on Eurostat for regional population data, which covers EU member states and candidate countries. Five OECD regions in non-EU countries lack population features as a result. This has minimal impact on the overall graph, as the affected regions represent a small fraction of the 444 total.

Finally, the novel predictions, both biological and environmental, have not been validated experimentally. The chemical-disease associations reflect structural patterns in the embedding space, not confirmed causal mechanisms. They should be treated as hypotheses for further investigation.

6.3 Future Work

Several directions for future research follow from the results and limitations described above.

The most impactful extension would be to validate the chemical-region predictions against independent data sources. The 6,661 novel predictions could be compared against measurement campaigns conducted after the training data cutoff, or against independent air quality models such as the Copernicus Atmosphere Monitoring Service (CAMS). Cities that the model predicts as high-exposure and that are subsequently confirmed by measurements would provide strong evidence that the learned representations capture real environmental patterns.

On the modelling side, alternative GNN architectures could address the receptive field limitation. Graph Attention Networks (GAT) or CompGCN could handle the 4-hop cross-domain path more effectively. Another option is to precompute cross-domain features by aggregating environmental statistics per country and attaching them directly to Disease nodes, bypassing the multi-hop path entirely. Recent GNN frameworks that support edge features natively (such as TransformerConv) would allow collapsing the CLA reification nodes into edges while preserving their quantitative values, shortening the cross-domain paths.

The chemical-disease associations could be investigated through targeted literature review. The model's finding that cadmium associates most strongly with Bronchioloalveolar Adenocarcinoma and arsenic with Adenocarcinoma of lung generates specific, testable hypotheses about subtype-specific carcinogenic mechanisms. Occupational health literature on cadmium and arsenic exposure in industrial settings could provide evidence for or against these associations.

The temporal analysis could be extended by incorporating more recent cancer registry data as it becomes available. The current model is trained on historical data up to 2024. Retraining with updated data would allow validation of the observed temporal trends and potentially enable short-term forecasting of cancer burden by region.

The chemical-region predictions could be operationalised as a monitoring prioritisation tool. Environmental agencies face resource constraints when deciding where to deploy measurement stations. A ranked list of regions where the model predicts unmeasured pollutant exposure could inform these allocation decisions. This would require packaging the prediction pipeline into a tool that can be updated as new EEA data becomes available.

Finally, the approach developed in this thesis is not specific to lung cancer. The pipeline architecture, the ontological alignment rules and the GNN evaluation framework could be applied to other disease domains where environmental and molecular data need to be integrated within a single knowledge graph.

6.4 Final Remarks

This thesis made three contributions to the LUCIA project. The first is a clean, structured and validated environmental dataset that replaces the problematic data that existed before this work. The four pipelines, the ontological rules and the shared entity framework provide a reusable foundation for future extensions. The second is a demonstrated link prediction capability on the full LUCIA knowledge graph, with R-GCN outperforming baseline models and producing biologically plausible gene-disease predictions. The third, and most significant, is the set of environmental link predictions that demonstrate how a trained GNN can extract actionable insights from environmental health data: which pollutants are most associated with which cancer subtypes, where unmeasured exposures likely exist and which countries show the strongest predicted associations over time.

The environmental data in the LUCIA knowledge graph is no longer a secondary annotation. It is a source of predictions that can guide monitoring decisions, generate hypotheses about subtype-specific carcinogenic mechanisms and identify populations at risk. The tools to extend and query this data are in place. The question is no longer whether environmental data belongs in a lung cancer knowledge graph, but how to best exploit the connections it creates.

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